

The Fast Mixing Mechanism for Differential Privacy

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Abstract

Randomized sketching is a central tool for compressing large-scale optimization problems while preserving accuracy. In particular, sketches that are based on structured matrices, such as the Hadamard matrix, can be applied efficiently and often yield solutions that approximate those of the original problem at much lower computational cost. In differential privacy (DP), Gaussian sketching has been used to solve DP linear regression, beginning with Sheffet [2017, 2019] and later refined by Lev et al. [2025, 2026]. However, although these methods achieve strong utility guarantees, they usually do not improve runtime over classical DP approaches. In this work, we introduce a new DP sketching mechanism based on fast transforms, which, in certain cases, matches the runtime of classical fast sketching methods. We prove state-of-the-art privacy guarantees for this mechanism and show that, in favorable regimes, they match those of the Gaussian sketch up to a constant factor. As an application, we combine this mechanism with recent sketch-based methods for DP linear regression to obtain a new algorithm with strong utility and improved runtime. We establish privacy and accuracy guarantees for this algorithm, yielding, to the best of our knowledge, the first fast method for DP ordinary least squares.

1 Introduction

Modern technologies to collect and process data have led to the development of large-scale datasets that require computationally efficient methods to analyze them. In recent years, *randomized sketching* has proven to be an attractive approach for designing computationally efficient data analysis tools; see Woodruff [2014] for an overview. Roughly speaking, these sketching methods transform the data via random projections, and the data analysis is conducted on this transformed data. These random projections are developed with two key goals in mind: (a) reducing the size of the data, and consequently the computational cost of the data analysis, and (b) preserving essential structures of the original data, therefore ensuring that the conclusion of the data analysis does not differ vastly from the original data analysis. This motivates the development of random projections that address both desiderata, and there has been significant progress made towards this recently [Drineas et al., 2006, Sarlos, 2006, Ailon and Liberty, 2009, 2013, Drineas et al., 2011, Pilanci and Wainwright, 2015, 2016, 2017, Cohen et al., 2016, Avron et al., 2017, Derezhinski et al., 2021].

Simultaneously, with the advent of these large datasets — particularly those built using personal data, there has been increased attention towards preserving the privacy of the participating individuals. Differential privacy (DP) [Dwork et al., 2006] provides a rigorous theoretical framework for the development of mechanisms that provably protect privacy, and DP mechanisms have been deployed

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in practice of late [Apple Research, 2017, Bureau, 2019, Facebook Research, 2020, Snap Security, 2022]. Here, the tradeoff to keep in mind is between preserving privacy and the quality of the data analysis as compared to the non-private data analysis; for instance, a completely garbled output achieves complete privacy, but fundamentally disregards the data analysis problem.

A relatively underexplored direction is to view the randomness of randomized sketching as a means of providing DP. A notable example is the work of Blocki et al. [2012], which proved that random projections can provide differential privacy, and the work of Sheffet [2017, 2019], which builds on this work and demonstrates how Gaussian sketching can be applied to private linear regression. More recent papers [Li et al., 2025, Lev et al., 2025, 2026] refine the privacy analysis for Gaussian sketching, develop improved algorithms for private linear regression, and demonstrate gains over standard baselines both theoretically and empirically. Yet these methods do not provide the main computational benefit usually associated with sketching. Specifically, if $S \in \mathbb{R}^{k \times n}$ is a dense Gaussian matrix and $X \in \mathbb{R}^{n \times d}$, computing SX costs $O(knd)$ operations, often comparable to solving the original problem. In contrast, modern non-private sketching methods are often built using sparse or structured transforms that are both computationally fast and preserve structures of the data, such as the subsampled randomized Hadamard transform (SRHT). Based on the above discussion, our work is motivated by the natural hypothesis:

Can we design a fast DP mechanism that preserves properties essential for data analysis?

This question was first studied by Upadhyay [2014], who proposed a DP sketching mechanism that can be applied significantly faster than the Gaussian sketch. However, the performance of their method is usually unsatisfactory; while their mechanism works in principle, we demonstrate that it is impractical for typical privacy regimes, incurring a substantial slack relative to the privacy guarantees of the Gaussian sketch. Thus, from a practical standpoint, this question remains open.

1.1 Our Contributions

We introduce a new family of fast DP sketches, which closes a gap between non-private fast sketching methods and recently studied private sketches. Our fast DP sketching mechanism (presented in Section 3) applies two sketches, first a large fast sketch that compresses the dataset, and second a small Gaussian sketch which provides DP. This approach is analogous to prior works on non-private fast sketching for approximate matrix multiplication (see Cohen et al. [2016, Appendix A.3] for example). For an input matrix $X \in \mathbb{R}^{n \times d}$ and a target sketch size k , this scheme is faster than the dense Gaussian sketch by a factor $\tilde{O}(\min\{\frac{n}{k}, k\})$. To show that this is indeed private, we provide Rényi-DP (a stronger notion of DP) guarantee for our proposed scheme.

We then apply our mechanism to DP ordinary least squares (DP-OLS). Instantiating it with SRHT sketches, we derive privacy and accuracy guarantees that explicitly quantify the price of fast, private sketching relative to non-fast constructions. For well-conditioned design matrices, these guarantees match those of state-of-the-art slow methods for DP-OLS. Experiments on large-scale linear regression datasets show that our method achieves state-of-the-art running times while maintaining accuracy close to non-fast baselines. In contrast, Upadhyay [2014] do not apply their fast sketching mechanism to DP-OLS, nor do they provide accuracy guarantees comparable to those of state-of-the-art DP-OLS methods. Thus, to our knowledge, our work provides the first fast algorithm for DP-OLS.

2 Problem Setup and Background

Notation. Random variables are in sans-serif (e.g., $\mathbf{X}, \mathbf{y}$), and their realizations in serif (e.g., X, y). The set $\{1, \dots, n\}$ is denoted by $[n]$. The ℓ_2 norm of $v \in \mathbb{R}^d$ is $\|v\|$. Let $X \in \mathbb{R}^{n \times d}$. We use $\lambda_{\min}^X, \lambda_{\max}^X, \|X\|_{\text{op}}$ to denote $\lambda_{\min}(X^\top X)$, $\lambda_{\max}(X^\top X)$, and $\sqrt{\lambda_{\max}^X}$ respectively. The all-zeros column vector in $\mathbb{R}^d$ is $\vec{0}_d$ and the $k \times k$ identity matrix is $\mathbb{I}_k$. The Gaussian distribution with mean μ and (co)variance Σ is $\mathcal{N}(\mu, \Sigma)$, and $\mathcal{N}(0, \mathbb{I}_{k_1 \times k_2})$ denotes a $k_1 \times k_2$ random matrix with i.i.d. standard normal entries. Throughout, $\tilde{O}(\cdot)$ hides polylogarithmic factors only in quantities whose dependence in the displayed bound is at most polynomial. The standard basis for $\mathbb{R}^d$ is denoted by $\{e_i\}_{i=1}^d$, where $e_i \in \mathbb{R}^d$ are the columns of $\mathbb{I}_d$. Additional notation is in Appendix A.

Problem Setup. We are interested in designing differentially private mechanisms that operate over tabular datasets. These are represented as a matrix $X \in \mathbb{R}^{n \times d}$. We make a bounded-domain assumption, namely, that there exists a known constant C_X such that $\|x_i\| \leq C_X$ for all $i \in [n]$, as in several prior analyses in both private and non-private settings (e.g., Shamir [2015], Wang [2018]). Specifically, from a privacy perspective, this condition can be enforced by “clipping” the data. Throughout, we assume that $C_X = 1$ without loss of generality. Since $X' = X/C_X$ and $\theta' = C_X \theta$ give $X\theta = X'\theta'$, any bound for $C_X = 1$ extends to general C_X by rescaling.

2.1 Differential Privacy

Differential privacy [Dwork et al., 2006] is a property of a randomized mechanism $\mathcal{M}$ that quantifies how different the output distributions of $\mathcal{M}$ are on “neighboring” datasets. Here, datasets X, X' are *neighbors* if X' is formed by removing an element from X , or vice versa.¹ We denote this relation by $X \simeq X'$. Throughout, we use the following standard definitions.

Definition 1 ((ε, δ) -DP). A randomized mechanism $\mathcal{M}$ is said to satisfy (ε, δ) -differential privacy if for all X, X' such that $X' \simeq X$ and measurable subsets $\mathcal{S} \subseteq \text{Range}(\mathcal{M})$,

$$\Pr(\mathcal{M}(X) \in \mathcal{S}) \leq e^\varepsilon \cdot \Pr(\mathcal{M}(X') \in \mathcal{S}) + \delta.$$

Our work leverages the Rényi-DP [Mironov, 2017] variant of differential privacy.

Definition 2 ($(\alpha, r(\alpha))$ -RDP). A randomized mechanism $\mathcal{M}$ is said to satisfy $(\alpha, r(\alpha))$ -RDP for some $\alpha > 1$ if for all X, X' such that $X \simeq X'$,

$$D_\alpha(\mathcal{M}(X) \parallel \mathcal{M}(X')) \leq r(\alpha),$$

where $D_\alpha(\mathbb{P} \parallel \mathbb{Q}) := \frac{1}{\alpha-1} \log(\mathbb{E}_{x \sim \mathbb{Q}} [\mathbb{P}^\alpha(x) \mathbb{Q}^{-\alpha}(x)])$ denotes the α -Rényi divergence [Rényi, 1960].²

One can translate $(\alpha, r(\alpha))$ -RDP to classical (ε, δ) -DP as demonstrated below.

Proposition 1 (Canonne et al. [2020, Proposition 12]). *If $\mathcal{M}$ satisfies $(\alpha, r(\alpha))$ -RDP, then it also satisfies (ε, δ) -DP for any $0 < \delta < 1$, where $\varepsilon = r(\alpha) + \log\left(1 - \frac{1}{\alpha}\right) - \frac{\log(\alpha\delta)}{(\alpha-1)}$.*

Both $(\alpha, r(\alpha))$ -RDP and (ε, δ) -DP are preserved under post-processing and degrade gracefully under composition. In particular, post-processing ensures that if a mechanism f satisfies either definition, then so does $g \circ f$ for any (possibly randomized) g [Dwork et al., 2014, Mironov, 2017].

¹For simplicity, we identify row removal with replacement by $\vec{0}_d$, so the dimension remains constant. This notion is sometimes called *zero-out* neighboring.

²Given random variables $\mathbf{p} \sim \mathbb{P}$ and $\mathbf{q} \sim \mathbb{Q}$, $D_\alpha(\mathbf{p} \parallel \mathbf{q})$ denotes the α -Rényi divergence $D_\alpha(\mathbb{P} \parallel \mathbb{Q})$.

2.2 Subsampled Randomized Hadamard Transform

Our fast DP sketch is based on the Subsampled Randomized Hadamard Transform (SRHT), which we discuss in this subsection. This is a standard tool for fast dimensionality-reduction of large matrices via matrix multiplication [Tropp, 2011, Ailon and Liberty, 2013].

Definition 3 (SRHT). Let n be a power of 2. An SRHT matrix $S_{\text{SRHT}} \in \mathbb{R}^{k \times n}$, $k \leq n$, is defined as

$$S_{\text{SRHT}} := \sqrt{\frac{n}{k}} P_{k \times n} H_n B_n$$

where $B_n = \text{diag}(\mathbf{b}_1, \dots, \mathbf{b}_n)$ and $\mathbf{b}_i$ are drawn independently from a Rademacher distribution, $H_n \in \mathbb{R}^{n \times n}$ is the normalized Hadamard matrix [Walsh, 1923] that satisfies $H_n^\top H_n = \mathbb{I}_n$, and $P_{k \times n}$ is formed by randomly sampling k rows of $\mathbb{I}_n$ uniformly without replacement.

We provide background on Hadamard matrices and properties of SRHT matrices in Appendix B. The SRHT is particularly appealing from a computational perspective; computing $S_{\text{SRHT}}X$ for $X \in \mathbb{R}^{n \times d}$ takes $O(nd \log(k))$ operations [Ailon and Liberty, 2009], which scales exponentially better in k than the cost of left multiplying X by an arbitrary, $k \times n$ dense matrix. Statistically, the SRHT also ensures that the minimal singular value and the norms of the rows are approximately preserved, while controlling the correlations between the columns of the output. These properties are based on prior results of Pilanci and Wainwright [2015] and Cohen et al. [2016], and are used in the analysis of our fast DP sketching mechanism. We discuss these in more detail in Appendix E.

Remark 1. The “power of 2” condition on n in the definition of SRHT is due to the standard recursive construction of Hadamard matrices, which applies under this condition. Other fast orthogonal transforms, such as DCT- or DFT-based constructions [Avron et al., 2010, Boutsidis and Gittens, 2013], remove this restriction, and can be used throughout to derive similar results. However, their unequal column norms require extra normalization in our privacy analysis, which degrades the signal strength.

2.3 Differential Privacy via Gaussian Sketches

As a prelude to our fast DP sketching mechanism, we first discuss the “slow” version which is based on a dense sketching matrix, specifically defined by $S_G \sim \mathcal{N}(0, \mathbb{I}_{k \times n})$ for a target sketch size k and input matrix $X \in \mathbb{R}^{n \times d}$. The use of such Gaussian sketches for information-theoretic privacy dates back to the work of Zhou et al. [2007], and Blocki et al. [2012] adapted it for differentially-private covariance estimation. Later work by Sheffet [2017, 2019] studies the Gaussian sketching mechanism in isolation; this analysis was recently sharpened by Lev et al. [2025], who provide a tighter Rényi differential privacy analysis of the mechanism. In the following lemma, we present the mechanism and its privacy guarantee from Lev et al. [2025].

Proposition 2 ([Lev et al., 2025]). Let $X \in \mathbb{R}^{n \times d}$ such that $\|x_i\| \leq 1$ for all $i \in [n]$ and let

$$\mathcal{M}(X) := \mathbf{g}^\top X + \sigma \xi^\top, \quad \text{with } \mathbf{g} \sim \mathcal{N}(\vec{0}_n, \mathbb{I}_n), \xi \sim \mathcal{N}(\vec{0}_d, \mathbb{I}_d).$$

Assume that we have knowledge of $\bar{\lambda}_{\min}$ such that $\lambda_{\min}^X \geq \bar{\lambda}_{\min}$, and that $\gamma := \sigma^2 + \bar{\lambda}_{\min} > 1$. Then, for every $\alpha \in (1, \gamma)$, $\mathcal{M}(X)$ is $(\alpha, \varphi(\alpha; \gamma))$ -RDP, where

$$\varphi(\alpha; \gamma) := \frac{\alpha}{2(\alpha - 1)} \log \left(1 - \frac{1}{\gamma} \right) - \frac{1}{2(\alpha - 1)} \log \left(1 - \frac{\alpha}{\gamma} \right). \quad (1)$$

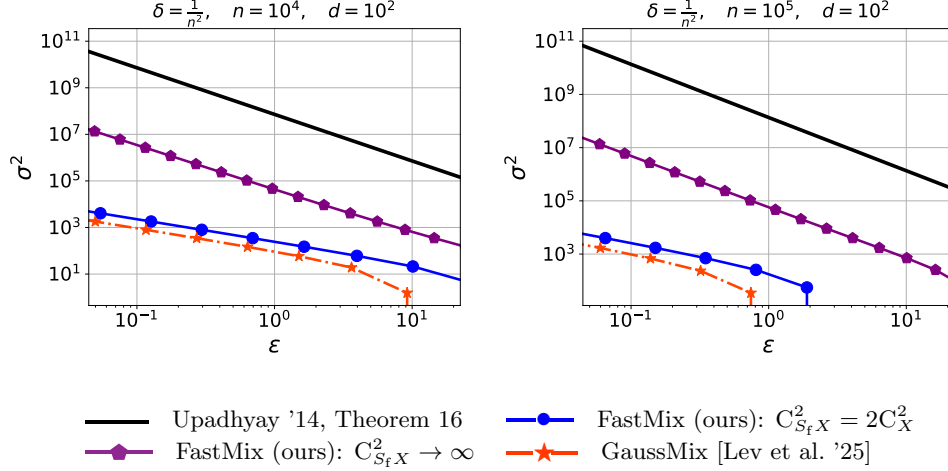


Figure 1: The additive noise scale σ required to attain target ε for Lemma 1 compared with Upadhyay [2014] and GaussMix [Lev et al., 2025, Lemma. 1] for $n \in \{10^4, 10^5\}$. We set $\lambda_{\min}^X = \frac{n}{8d}C_X^2$ and $\lambda_{\min}^{S_f X} = \frac{5}{6}\lambda_{\min}^X$. We plot the noise required by FastMix with $C_{S_f X}^2 = 2C_X^2$, and in the extreme case $C_{S_f X}^2 \rightarrow \infty$. The curves use the exact characterization instead of (3) (see Appendix C for details).

Proposition 2 bounds the divergence incurred by multiplying X by a single Gaussian vector. For a Gaussian sketch S_G with k independent Gaussian rows, each row corresponds to one such release; hence, by composition in RDP, the divergence bound for $S_G X$ is k times the bound in (1). To interpret the role of γ , note that it can be viewed as a lower bound on the squared minimal singular value of the augmented matrix $\tilde{X} := [X^\top, \sigma \mathbb{I}_d]^\top$. Thus, given any lower bound $\lambda_{\min}^X \geq \bar{\lambda}_{\min}$, the term $\sigma\xi$ can be interpreted as lifting the minimal singular value to a desired level γ . Gaussian sketches can yield state-of-the-art accuracy guarantees for DP linear regression [Lev et al., 2026], improving over classical DP constructions such as Wang [2018]; however, they do not improve the asymptotic runtime. Indeed, forming $S_G X$ costs $O(nkd)$ in the worst case. Since typically $k = \Omega(d)$, this is at least $O(nd^2)$, which is often comparable to solving the underlying optimization problem.

3 The Fast Mixing Mechanism for Differential Privacy

In this section, we present our key development: a DP mechanism with worst-case runtime matching SRHT for sufficiently large n . It builds on the simpler mechanism

$$\mathcal{M}_{S_f}(X) = \mathbf{g}^\top S_f X + \sigma \xi^\top$$

where $S_f \in \mathbb{R}^{k \times n}$ is a prespecified matrix, $\mathbf{g} \sim \mathcal{N}(\vec{0}_k, \mathbb{I}_k)$, $\xi \sim \mathcal{N}(\vec{0}_d, \mathbb{I}_d)$, and $\sigma > 0$. The following lemma proves the RDP guarantees of $\mathcal{M}_{S_f}$, generalizing the result of Lev et al. [2025, Lemma 1].

Lemma 1. *Let X be a dataset such that $\|x_i\| \leq 1$ for all $i \in [n]$. Suppose $S_f \in \mathbb{R}^{k \times n}$ is a matrix such that $\|S_f e_i\| \leq 1$ for all $i \in [n]$. Define $C_{S_f X}^2 := 1 + 2 \cdot \max_{i \in [n]} \|X^\top S_f^\top S_f e_i - x_i\|$. Assume that*

$$\gamma := C_{S_f X}^{-2} \cdot \left(\lambda_{\min}^{S_f X} + \sigma^2 \right) > \frac{25}{8}. \quad (2)$$

Then

$$\max_{X' \sim X} D_\alpha(\mathcal{M}_{S_f}(X) \parallel \mathcal{M}_{S_f}(X')) \leq \frac{25\alpha}{32\gamma^2}, \quad \text{for all } \alpha \in \left(1, \frac{8\gamma}{25}\right). \quad (3)$$

Algorithm 1 FastMix

Input: Dataset $X \in \mathbb{R}^{n \times d}$, parameters $k_1, \gamma, \tau, \omega$, sketch $S_f \in \mathbb{R}^{k_2 \times n}$.

- 1: $Z \leftarrow S_f X, \hat{m} \leftarrow \max_{i \in [n]} \|Z^\top S_f e_i - x_i\|, \hat{\Delta} \leftarrow \max_{i, j \in [n]} |e_i^\top (S_f^\top S_f - \mathbb{I}_n) e_j|$ and $\hat{\Lambda} \leftarrow 2 - \min_{i \in [n]} \|S_f e_i\|^2$.
 - 2: $\tilde{m} \leftarrow \max \left\{ 0, \hat{m} + \omega \hat{\Delta} (\tau - z_1) \right\}$ for $z_1 \sim \text{Lap}(0, 1)$.
 - 3: $\tilde{\lambda} \leftarrow \max \left\{ 0, \lambda_{\min}^Z - \omega \left(\hat{\Lambda} + 2\tilde{m} \right) (\tau - z_2) \right\}$ for $z_2 \sim \text{Lap}(0, 1)$.
 - 4: $\tilde{\eta} \leftarrow \sqrt{\max \left\{ 0, \gamma (1 + 2\tilde{m}) - \tilde{\lambda} \right\}}$.
 - 5: **Output:** $S_G Z + \tilde{\eta} \xi$ with $S_G \sim \mathcal{N}(0, \mathbb{I}_{k_1 \times k_2})$, $\xi \sim \mathcal{N}(0, \mathbb{I}_{k_1 \times d})$, and $\tilde{\eta}$.
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Proof Sketch The proof follows by calculating the α -Rényi divergence between $\mathcal{M}_{S_f}(X)$ and $\mathcal{M}_{S_f}(X')$. This is possible since $\mathbf{g}^\top S_f X$ and $\mathbf{g}^\top S_f X'$ are both zero mean Gaussian vectors with covariances $X^\top S_f^\top S_f X$ and $X'^\top S_f^\top S_f X'$. The proof then follows by using the closed-form formula for the Rényi divergence between multivariate Gaussian vectors, and by crucially noting that since $X' \simeq X$, the covariances differ by a rank-2 update. We then use the assumption made on γ in (2) to derive an upper bound that is independent of X . The full proof is presented in Appendix C.1.

The quantity $C_{S_f X}$ can be interpreted as the row bound for $S_f X$. Similarly to the Gaussian sketching mechanism, for every $\alpha > 1$ and fixed $C_{S_f X}$, this upper bound decays to zero as γ increases. Thus, (3) can be made arbitrarily small by increasing γ , which in turn is done by increasing the noise level σ^2 . Thus, γ parallels a similar quantity in Lev et al. [2025, Lemma. 1], calculated for the matrix $S_f X$ instead of X . The mechanism $\mathcal{M}_{S_f}$ can be viewed as applying a premultiplication to the augmented matrix $[(S_f X)^\top, \sigma \mathbb{I}_d]^\top$ by a Gaussian random vector of length $k + d$. We also recover the result of Lev et al. [2025, Lemma. 1] when $S_f^\top S_f = \mathbb{I}_n$, in which case $\mathcal{M}_{S_f}$ coincides with $\mathcal{M}$ in distribution.

The simpler mechanism $\mathcal{M}_{S_f}$ is the basis for our proposed fast mixing mechanism FastMix which we discuss next. While (2) provides guidance for choosing σ , γ is not a private quantity by definition. A key step in FastMix is in selecting σ for $\mathcal{M}_{S_f}$ privately, which is achieved by private estimation of $C_{S_f X}^2$ and $\lambda_{\min}^{S_f X}$. We do so by identifying the sensitivity (i.e., the maximal change for neighboring datasets) of $X \mapsto C_{S_f X}^2$ and $X \mapsto \lambda_{\min}^{S_f X}$ and apply the Laplace mechanism to ensure privacy (Lines. 2 and 3). After this, we calibrate the noise level with these private estimates and apply $\mathcal{M}_{S_f}$.

We recapitulate our goals: we are interested in designing a mechanism that is both fast and has a comparable noise budget to the dense Gaussian sketch. The latter can be satisfied if the transform $X \mapsto S_f X$ approximately preserves $\lambda_{\min}^X$ and C_X^2 . Curiously, we find that SRHT provides a means to achieve both of these goals. First, as discussed a priori, the SRHT is faster than premultiplying X by a dense Gaussian matrix. Second, for any $\chi \in (0, 1]$, when the target sketch size k satisfies $k\chi^2 = \Theta(\text{rank}(X) \cdot \log^4(n))$ implies that

$$\left| \lambda_{\min}^{S_f X} - \lambda_{\min}^X \right| \leq \chi \cdot \lambda_{\min}^X \quad \text{and} \quad C_{S_f X}^2 - C_X^2 \leq 2\chi \cdot C_X \|X\|_{\text{op}}.$$

See Appendix E for an extended discussion of this. Our main theorem for FastMix is stated next.

Theorem 1. *Let $\delta \in (0, 1)$, $k_1 \geq 1$ and S_f matrix satisfying $\|S_f e_i\| \leq 1$ for all $i \in [n]$. Then,*

1. *For every $\gamma > 25/8$ and $\omega > 0$, the output of Algorithm 1 with $\tau \geq \log(3/2\delta)$ is $(\varepsilon(\omega, \gamma, k_1, \delta), \delta)$ -DP with*

respect to X where

$$\varepsilon(\omega, \gamma, k_1, \delta) \leq \frac{2}{\omega} + \begin{cases} \frac{25k_1}{32\gamma^2} + 2\sqrt{\frac{25k_1 \log(3/\delta)}{32\gamma^2}}, & \log(3/\delta) \leq \left(\frac{8\gamma}{25} - 1\right)^2 \cdot \frac{25k_1}{32\gamma^2}, \\ \frac{k_1}{4\gamma} + \frac{25 \log(3/\delta)}{8\gamma - 25}, & \text{otherwise.} \end{cases}$$

2. For $S_f \in \text{SRHT}(k_2, n)$, Algorithm 1 runs in $O(nd \log(n) + k_1 k_2 d + k_2 d^2 + d^3)$.

Proof Sketch We note that **FastMix** comprises three private-mechanism calls: calculating $\tilde{m}$, calculating $\tilde{\lambda}$, and computing the output $S_G Z + \tilde{\eta} \xi$. First, note that $\tilde{m} \geq \hat{m}$ and $\tilde{\lambda} \leq \lambda_{\min}^Z$ w.p. at least $1 - 2\delta/3$, which holds since $\tau \geq \log(3/2\delta)$ and since $\mathbf{z}_1$ and $\mathbf{z}_2$ are Laplace variables. Conditioned on this event, note that the sensitivity of $\tilde{m}$ is upper bounded by $\hat{\Delta}$ (Lemma 4) and the sensitivity of $\lambda_{\min}^Z$ is upper bounded by $\hat{\Lambda} + 2\hat{m} \leq \hat{\Lambda} + 2\tilde{m}$ (Lemma 5). Thus, the privacy properties of the Laplace mechanism apply to the release of $\tilde{m}$, while for the release of $\tilde{\lambda}$, the privacy guarantees hold since we use a private upper bound on the local sensitivity. Then, since S_f has unit column-norms, the conditions of Lemma 1 hold. Multiplying the guarantees of Lemma 1 by k_1 , converting to (ε, δ) -DP using Proposition 1, and combining with the guarantees of the steps that use the Laplace mechanism yields the overall privacy guarantees. The runtime argument follows using the fact that the runtime complexity of sketching with an SRHT costs $O(nd \log(n))$, and using properties of the Hadamard matrix for computing the different quantities efficiently. The full proof, along with the tightest bound on ε obtained by using closed-form guarantees obtained for Lemma 1, is in Appendix C.4.

Remark 2. **FastMix** uses the Laplace mechanism to compute $\tilde{m}$ and $\tilde{\lambda}$, for slightly better empirical performance. An analogous construction can be obtained with the Gaussian mechanism.

Role of S_f . The matrix S_f is treated as deterministic, and our privacy analysis does not rely on any randomness in its construction. The quantity $\hat{\Delta} = \max_{i \neq j} |e_i^\top S_f^\top S_f e_j|$ is referred to as the *coherence* of S_f . The application of S_f can be viewed as a computationally efficient pre-processing step that also “accelerates” the application of the Gaussian sketch. Since S_f is assumed to be public, any quantity that depends only on S_f (such as $\hat{\Delta}$ and $\hat{\Lambda}$) need not itself be privatized. As discussed previously, the SRHT is viable from a computational and privacy standpoint, and additionally satisfies $\hat{\Lambda} = 1$. Additionally, for the SRHT, we can rely on classical sketching results to obtain high-probability approximation guarantees for downstream applications, as shown in Section 4.

Analytical Bound. As shown in Appendix D, Lemma 1 implies $\varepsilon = O\left(\frac{C_{S_f X}^4 k_1}{(\lambda_{\min}^{S_f X} + \sigma^2)^2} + \frac{C_{S_f X}^2 \sqrt{k_1 \log(1/\delta)}}{\lambda_{\min}^{S_f X} + \sigma^2}\right)$ whenever $\lambda_{\min}^{S_f X} + \sigma^2 > a_0$ and $k_1 > a_1 \cdot \log(1/\delta)$ for constants a_0 and a_1 . Moreover, for any $\omega \geq \frac{3}{\varepsilon}$, the releases $\tilde{\lambda}$ and $\tilde{m}$ contribute at most $\frac{2\varepsilon}{3}$ to ε , so **FastMix** matches this bound up to constant degradation in ε . In comparison, the pure Gaussian sketch with k_1 rows satisfies $\varepsilon = O\left(\frac{k_1}{(\lambda_{\min}^X + \sigma^2)^2} + \frac{\sqrt{k_1 \log(1/\delta)}}{\lambda_{\min}^X + \sigma^2}\right)$ [Lev et al., 2025, Section 4]. Thus, **FastMix** incurs two degradations: each factor $\frac{\sqrt{k_1}}{\lambda_{\min}^X + \sigma^2}$ is multiplied by $C_{S_f X}^2 \geq 1$, increasing ε ; and $\lambda_{\min}^X$ is replaced by $\lambda_{\min}^{S_f X}$. As discussed, the latter typically changes $\lambda_{\min}^X$ only by a constant multiplicative factor, whereas the former can be as large as $O\left(\sqrt{\frac{\text{rank}(X) \log^4(n)}{k_2}} \cdot n\right)$.

Algorithm 2 Fast IHM

Input: Dataset (X, Y) ; noise parameters $(\gamma, \tau, \sigma, \omega)$; sketch sizes (k_1, k_2) ; iterations T ; clip $C > 0$.

- 1: Initialize $\hat{\theta}_0 \leftarrow \vec{0}_d$.
 - 2: Compute $\eta, \{\tilde{X}_t\}_{t=0}^{T-1} \leftarrow \text{MultipleFastMixing}(X, 1, \gamma, \tau, \omega, k_2, T)$.
 - 3: **for** $t = 0, \dots, T-1$: **do**
 - 4: Sample $S_{G,t} \sim \mathcal{N}(0, \mathbb{I}_{k_1 \times k_2})$, $\xi_t \sim \mathcal{N}(0, \mathbb{I}_{k_1 \times d})$, $\zeta_t \sim \mathcal{N}(\vec{0}_d, \mathbb{I}_d)$.
 - 5: Calculate $\hat{X}_t = S_{G,t} \tilde{X}_t + \eta \xi_t$.
 - 6: Calculate $\tilde{G}_t = X^\top \text{clip}_C(Y - X \hat{\theta}_t) - \eta^2 \hat{\theta}_t + \sigma \zeta_t$.
 - 7: Update $\hat{\theta}_{t+1} = \hat{\theta}_t + (\frac{1}{k_1} \hat{X}_t^\top \hat{X}_t)^{-1} \tilde{G}_t$.
 - 8: **Output:** $\hat{\theta}_T$
-

Comparison to Existing Literature. To our knowledge, little prior work has studied DP via fast sketches. The closest related mechanism is due to Upadhyay [2014], whose construction also first applies an SRHT, but then uses a sparse Gaussian outer sketch. Their privacy bound scales as $O\left(\log(1/\delta) \sqrt{\frac{k \log(1/\delta)}{\lambda_{\min}^{S_f X} + \sigma^2}}\right)$, where S_f is the SRHT. This is looser than FastMix by a factor $O\left(\frac{\log(1/\delta) (\lambda_{\min}^{S_f X} + \sigma^2)^{1/2}}{C_{S_f X}^2}\right)$. Their guarantee resembles our analysis in Appendix C.2, corresponding to $C_{S_f X}^2 \rightarrow \infty$, with further constant and $\log(1/\delta)$ loss from analyzing directly in (ϵ, δ) -DP rather than via RDP. By Figure 1, our approach typically reduces σ^2 by more than two orders of magnitude.

Choosing k_2 . While Theorem 1 does not depend explicitly on k_2 , it depends on k_2 implicitly through $\lambda_{\min}^Z, \hat{m}$ and $\hat{\Delta}$. Note that, for matching the guarantee of the Gaussian sketch, our goal is to make $\lambda_{\min}^Z \approx \lambda_{\min}^X$ and to keep $\hat{m}$ and $\hat{\Delta}$ small. Without additional assumptions, choosing $S_f \sim \text{SRHT}(k_2, n)$ yields that for any $\chi \in (0, 1]$ s.t. $\frac{k_2 \chi^2}{\log^4(n)} \geq \max\{c_0 \cdot \text{rank}(X), c_2 \cdot \log(c_1/\epsilon)\}$, $\lambda_{\min}^Z \geq (1-\chi) \lambda_{\min}^X$ and $\hat{m} \leq \frac{\chi}{2} \|X\|_{\text{op}}$ w.p. at least $1-\epsilon$ for constants c_0, c_1, c_2 (Lemma 10). Moreover, if $k_2 \chi^2 \geq \tilde{c}_0 \log(\tilde{c}_1 n^2 / \chi \epsilon) \log(\tilde{c}_2 n^2 / \epsilon)$ then $\hat{\Delta} \leq \chi$ w.p. at least $1-\epsilon$ for another three constants $(\tilde{c}_0, \tilde{c}_1, \tilde{c}_2)$ (Lemma 9). Consequently, choosing k_2 above these values ensures that $\lambda_{\min}^Z$ and $C_{S_f X}^2$ are close to $\lambda_{\min}^X$ and C_X^2 up to factors $1 + \chi$ and $1 + \frac{\chi}{2C_X} \|X\|_{\text{op}}$, respectively, making the degradation relative to Proposition 2 bounded and allowing control over the privacy guarantee of FastMix.

Complexity. Setting $k_1 \chi^2 = O(\max\{d, \log(1/\epsilon)\})$ and $k_2 \chi^2 = \tilde{O}(\max\{\text{rank}(X), \log^2(1/\epsilon)\})$ yields runtime of $\tilde{O}\left(nd + d^3 + \frac{d}{\chi^4} \cdot \max\{\text{rank}(X), \log^2(1/\epsilon)\} \cdot \max\{d, \log(1/\epsilon)\}\right)$. As discussed, these values allow controlled degradation relative to Proposition 2, and thus represent typical choices of k_1 and k_2 . Whenever the dominating term is nd —corresponding to the regime where sketching is most beneficial—this matches, up to logarithmic factors, the cost of sketching with SRHT.

4 Application: Fast DP-OLS

We now present an application of FastMix to DP-OLS. Using FastMix, we obtain a new algorithm with state-of-the-art runtimes. We demonstrate empirical runtime improvements of factors up to ≈ 3 for large-scale datasets. As we show, in certain cases, these runtime improvements come without

paying for accuracy. Recall that the non-private OLS baseline is given by

$$\theta^* := \operatorname{argmin}_{\theta} \|Y - X\theta\|^2$$

where $X \in \mathbb{R}^{n \times d}$ and $Y \in \mathbb{R}^n$. Its DP version seeks a regressor $\hat{\theta}$ that satisfies (ε, δ) -DP (Definition 1) while minimizing $\|Y - X\hat{\theta}\|^2$. We focus on the regime $n \geq d$. To obtain a private and fast solution, we adapt the iterative Hessian mixing (IHM) algorithm of [Lev et al., 2026] to use **FastMix**. Our algorithm is presented in Algorithm 2; the next theorem provides its privacy and accuracy guarantees.

Theorem 2. *Let $\hat{\theta}$ be the output of Algorithm 2. Then, there exist $(\gamma, \tau, \sigma, \omega)$ such that*

1. *The output $\hat{\theta}$ is (ε, δ) -DP with respect to (X, Y) .*
2. *Let $\varrho \in (0, 1]$. Then, for every $\chi \in (0, \frac{1}{4}]$ there exist k_1, k_2, T , and C s.t. w.p. at least $1 - \varrho$:*

$$\|Y - X\hat{\theta}_T\|^2 - \|Y - X\theta^*\|^2 \leq \begin{cases} \tilde{O}(\gamma_f \cdot C_f^2(1 + \bar{m}) - \lambda_{\min}^X C_f^2) & \text{if } \lambda_{\min}^X \leq \gamma_f \\ \tilde{O}\left(\frac{\gamma_f^2 C_f^2}{\lambda_{\min}^X(1 + \bar{m})^2}\right) & \text{otherwise} \end{cases} \quad (4)$$

where

$$g^X := \left(\frac{\varepsilon d}{\sqrt{\log(1/\delta)}} (\lambda_{\max}^X - \lambda_{\min}^X) \right)^{2/3}, \quad C_f^2 := C_Y^2 + \|\theta^*\|^2, \quad \bar{m} := \chi \left(\sqrt{\lambda_{\max}^X} + \frac{\log(1/\delta)}{\varepsilon} \right) \quad (5)$$

$$\text{and } \gamma_f := \frac{\sqrt{\max\{d, \log(1/\varrho), g^X\} \log(1/\delta)(1 + \bar{m})}}{\varepsilon}.$$

3. *Under the previous $\chi, \varrho, k_1, k_2, T$ and C , the algorithm runs in*

$$\tilde{O} \left(nd + \frac{d}{\chi^2} \cdot \max\{d, g^X, \log(1/\varrho)\} \cdot \left(d + \frac{1}{\chi^2} \cdot \max\{\operatorname{rank}(X), g^X, \log^2(1/\varrho)\} \right) \right).$$

Proof Sketch The proof adapts the analysis of Lev et al. [2026] to **FastMix**, where the effective sketch concatenates two sketches. Using properties of SRHT, this roughly incurs a loss of factor $1 - \chi$ in $\lambda_{\min}^X$ and $1 + \bar{m}$ in γ . Substituting these in the asymptotic analysis of Lev et al. [2026] yields the stated argument. The runtime complexity is dominated by T steps of **FastMix** (splitted between **MultipleFastMixing** and Line. 5), and T iterations of solving an OLS problem of size $k_1 \times d$. Summing these and substituting the asymptotic arguments of the parameters k_1, k_2, C and T used for deriving (5) yields the stated argument. The full proof is in Appendix E.

The guarantee (4) parallels Lev et al. [2026, Theorem. 1]. The main difference is that the bound is multiplied by $1 + \bar{m}$ whenever $\lambda_{\min}^X \leq \gamma_f$, and that the threshold γ_f is inflated by the same factor. This is since the Gaussian sketch is applied to $\{S_{H,t}X\}_t$ rather than directly to X . This degradation can be controlled by increasing k_2 at the cost of a runtime increase, as demonstrated in Section 4.1. For runtime, whenever $d\chi^2 \geq \max\{\operatorname{rank}(X), g^X, \log^2(1/\varrho)\}$, the overall runtime evaluates to $\tilde{O}\left(nd + \frac{d^3}{\chi^2}\right)$, incurring a slight increase relative to the runtime of **FastMix** and shows the trade-off in χ . In particular, in many regimes, it is significantly better than the complexity of classical OLS solvers, given by $O(nd^2)$. Comparing this to the complexity of solving OLS using sketching with SRHT, which is $\tilde{O}\left(nd + \frac{d^2 \cdot \operatorname{rank}(X)}{\chi^2}\right)$ [Pilanci and Wainwright, 2015, Section. III.A], this complexity is worse whenever $\operatorname{rank}(X) \ll d$, and asymptotically matches whenever $n = \Omega\left(\frac{d^2}{\chi^2}\right)$.

Remark. Another fast DP-OLS algorithm can be obtained by combining **FastMix** with the sketch-and-solve method of **Sheffet** [2017]. However, sketch-and-solve sketches (X, Y) rather than X alone. Since $\lambda_{\min}^{(X,Y)} \leq \lambda_{\min}^X$ and $\lambda_{\max}^{(X,Y)} \geq \lambda_{\max}^X$,³ the resulting utility degradation is typically worse than in the IHM-based construction and, for practical ε , yields substantially lower accuracy. Thus, the practical gain comes not from **FastMix** alone, but from using it within IHM [Lev et al., 2026].

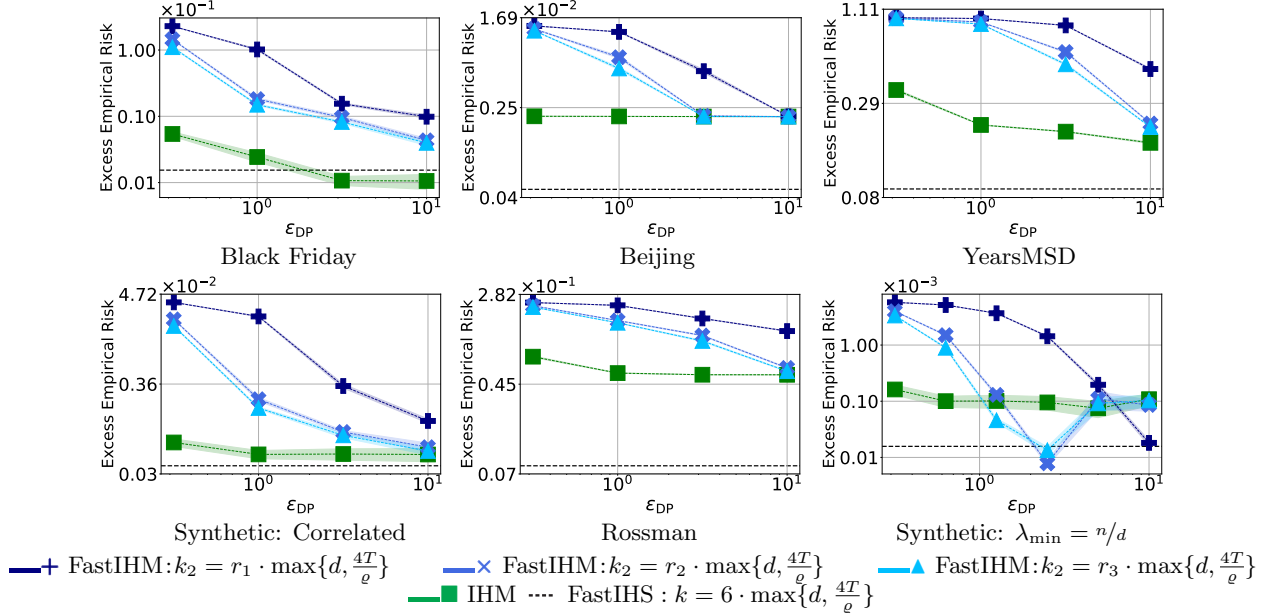


Figure 2: Performance comparison of **FastIHM** and **IHM**. The upper plots are for $\frac{n}{d} \approx 1450$ and the lower for $\frac{n}{d} \approx 16000$. For each dataset, we use three values of k_2 , represented by r_1, r_2 , and r_3 . The plots illustrate the dependence on $\lambda_{\max}^X$ and $\lambda_{\min}^X$, where smaller $\lambda_{\max}^X$ and larger $\lambda_{\min}^X$ improve performance. Moreover, in certain cases, **FastIHM** matches the performance of **IHM**.

Dataset	(r_1, r_2, r_3)	Fast IHS	Fast IHM: r_1	Fast IHM: r_2	Fast IHM: r_3
Black Friday	(4, 100, 200)	4.22	2.44	1.95	1.61
Beijing	(4, 100, 200)	6.89	3.12	2.38	1.94
YearsMSD	(4, 100, 200)	7.10	2.51	1.70	1.53
Synthetic Correlated	(4, 400, 800)	4.31	2.33	2.14	1.91
Rossmann	(4, 400, 800)	5.13	3.01	2.63	2.40
Synthetic: $\lambda_{\min} = \frac{n}{d}$	(4, 400, 800)	5.00	2.52	2.26	2.06

Figure 3: The ratio between the runtime of **IHM** and each of the different methods (**FastIHS** and **FastIHM**), for the three ratios r_1, r_2 and r_3 . In most cases, **FastIHM** is 2 to 3 times faster than **IHM**.

4.1 Simulations

Our experiments are designed to highlight the runtime advantage of **FastMix**. For accuracy comparisons, we use the **IHM** from Lev et al. [2026]. This method was shown to attain state-of-the-art performance over a large collection of datasets, and further fixes the underlying implementation, making the runtime differences only due to the replacement of the Gaussian sketch with **FastMix**.

³This holds since $\lambda_{\max}^{(X,Y)} = \max_v v^\top (X,Y)^\top (X,Y) v \geq \max_u u^\top X^\top X u = \lambda_{\max}^X$ and similarly for $\lambda_{\min}^{(X,Y)} \leq \lambda_{\min}^X$.

Our goal is to show that **FastMix** attains state-of-the-art runtime for a given approximation accuracy, rather than to improve on the best known accuracy guarantees, which were already established in [Lev et al., 2026]. We use IHS with SRHT (also termed *FastIHS*) as our non-private baseline. We evaluate the methods on six large-scale datasets chosen to illustrate the applicability of the utility guarantee in (4). Dataset statistics are reported in Appendix J, with full experimental details in Appendix H. In Appendix I, we also include experiments with extreme n/d , a regime where sketching is especially favorable. Each experiment shows the accuracy-privacy trade-off induced by varying k_2 , and we further report the speedup factor attained relative to IHM. As shown in Figure 3, **FastIHM** is substantially faster than IHM, with an accuracy loss that decreases as k_2 increases. The first row, corresponding to Black Friday, Years, and Beijing, considers datasets with $n/d \approx 1450$, $\lambda_{\min}^X \approx 0$, and decreasing $\lambda_{\max}^X$. When $\lambda_{\max}^X$ is smaller, as is the case for the Beijing and the YearsMSD datasets, **FastIHM** attains performance closer to IHM for the higher regime of ε , while remaining faster. For the Black Friday dataset, where $\lambda_{\max}^X \approx 4 \cdot 10^4$, there is a performance loss between **FastIHM** and IHM, even for larger ε 's. The second row considers datasets with $n/d \approx 2^{14}$ and varying values of $\lambda_{\min}^X$ and $\lambda_{\max}^X$. Synthetic Correlated and Rossman both have poorly conditioned design matrices; However, the final plot corresponds to a setting where $\lambda_{\min}^X$ nearly attains its maximal theoretical value, n/d . In this case, **FastIHM** matches IHM's accuracy, as predicted by (4). Surprisingly, in this setting, the performance of **FastIHM** is not monotone in ε : for some values of ε , it even outperforms IHM and matches the non-private baseline. We attribute this behavior to the implicit regularization induced by adding noise to the sketched Hessian. In IHM, the added noise level is always zero for all ε . In contrast, for **FastIHM**, the noise is zero only for the two largest values of ε and for the choices r_2 and r_3 . For smaller values of ε , the noise acts as a form of Hessian regularization. This is consistent with prior work showing that IHS can benefit from regularization or debiasing of the sketched Hessian [Zhang and Pilanci, 2023, Derezhinski et al., 2020]. In our setting, this effect arises implicitly through the privacy noise added to the sketched Hessian. Developing a theory for optimal Hessian debiasing in the private setting is an interesting direction for future work.

5 Discussion and Future Work

We introduced a fast DP mechanism based on sketching, paralleling fast sketching methods. The mechanism attains state-of-the-art DP guarantees and, to the best of our knowledge, is the first fast sketching-based DP mechanism whose privacy degradation relative to a Gaussian sketch is only a constant factor in some regimes. We applied it to DP-OLS, obtaining an algorithm with state-of-the-art runtime and provable accuracy guarantees. Relative to prior DP sketching-based approaches, our method is significantly faster and, in some regimes, incurs negligible accuracy degradation. Large-scale regression simulations confirm the gains, demonstrating speedups.

Our results suggest several directions for future work. The current instantiation of **FastMix** concatenates an SRHT with a Gaussian sketch. This choice has three drawbacks: it does not exploit sparsity in the input matrix; it does not minimize correlations between sketch columns, leaving $\hat{\Delta}$ in Line 1 relatively large (see Appendix F); and its cost depends on $k_1 k_2 d$, rather than kd^2 for a single sketch with k rows, slightly increasing the complexity relative to Pilanci and Wainwright [2015, Section. III.A], in particular whenever X has a low-rank structure. It would therefore be valuable to study alternative sketches within **FastMix**, such as sparse sketches or sketches with nearly minimal coherence, and to examine whether the concatenated sketch can be replaced by a single sketch. Such variants may improve both accuracy and runtime. In particular, since modern sketching methods adapt the sketch to the matrix being sketched (see, e.g., [Derezhinski et al., 2021]), it is interesting to

ask whether analogous constructions are possible under DP. Beyond DP-OLS, [FastMix](#) may serve as a primitive for second-order DP convex optimization algorithms in the spirit of the Newton sketch [[Pilanci and Wainwright, 2017](#)], enabling computationally efficient DP second-order methods.

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A Additional notation

Random variables are denoted using sans-serif fonts (e.g., $\mathbf{X}, \mathbf{y}$), while their realizations are represented by regular italics (e.g., X, y). The natural logarithm (with base e) is denoted by $\log(\cdot)$. The L_2 norm of a vector $v := (v_1, \dots, v_d)$ is given by $\left(\sum_{i=1}^d v_i^2\right)^{\frac{1}{2}}$ and is denoted by $\|v\|$, its L_∞ norm is given by $\max_{i=1, \dots, d} |v_i|$ and is denoted by $\|v\|_\infty$, and its Mahalanobis norm with respect to a PSD matrix B , namely, the quantity $(v^\top B v)^{\frac{1}{2}}$, is denoted by $\|v\|_B$. In the case where A is a square matrix, its trace and determinant are denoted by $\text{Tr}(A)$ and $\det(A)$ respectively, and its minimal and the maximal eigenvalues are $\lambda_{\min}(A)$ and $\lambda_{\max}(A)$ respectively. The minimal and maximal eigenvalues of its Gram matrix are denoted by $\lambda_{\min}^A := \lambda_{\min}(A^\top A)$ and $\lambda_{\max}^A := \lambda_{\max}(A^\top A)$. The operator norm of a matrix A is denoted by $\|A\|_{\text{op}}$ and is defined by $\sup_{v: \|v\|=1} \|Av\| = \sqrt{\lambda_{\max}^A}$ and the Frobenius norm by

$\|A\|_{\text{fro}} := \left(\sum_{i=1}^n \sum_{j=1}^d a_{ij}^2\right)^{1/2}$. We use the notation $A \succeq 0$ and $A \succ 0$ for denoting situations in which A is Positive Semi-Definite (PSD) and in which A is Positive Definite (PD). The Loewner order is defined in the usual way, where $B \preceq A$ denotes $A - B \succeq 0$. We usually denote our dataset $\{x_i\}_{i=1}^n$ where each $x_i \in \mathbb{R}^d$ in the matrix form $X = (x_1, x_2, \dots, x_n)^\top$. The set of integer numbers from 1 to n is denoted by $[n]$. The all zeros column vector of size d is denoted by $\vec{0}_d := (0, 0, \dots, 0)^\top$, and a matrix of size $a \times b$ full of zeros is denoted by $0_{a \times b}$. We denote by $\mathcal{N}(0, \mathbb{I}_{k_1 \times k_2})$ a $k_1 \times k_2$ matrix consisting of i.i.d. Gaussian elements with zero mean and unit variance. The $k \times k$ identity matrix is denoted by $\mathbb{I}_k$. We write $y(x) = O(x)$ if $\exists c > 0, x_0 > 0$ such that $|y(x)| \leq cx$ for all $x \geq x_0$. We write $y(x) = o(x)$ if $\lim_{x \rightarrow \infty} \frac{y(x)}{x} = 0$. We write $y(x) = \Omega(x)$ if $\exists c > 0, x_0 > 0$ such that $y(x) \geq cx$ for all $x \geq x_0$. We write $y(x) = \Theta(x)$ if $y(x) = O(x)$ and $y(x) = \Omega(x)$. Given two probability distributions $\mathbb{P}$ and $\mathbb{Q}$ and a scalar parameter $\lambda \in [0, 1]$, the notation $\lambda\mathbb{P} + (1 - \lambda)\mathbb{Q}$ denotes the distribution of the random variable $\mathbf{x} = \mathbf{b}\mathbf{p} + (1 - \mathbf{b})\mathbf{q}$ for $\mathbf{p} \sim \mathbb{P}, \mathbf{q} \sim \mathbb{Q}, \mathbf{b} \sim \text{Ber}(\lambda)$ and $\mathbf{p} \perp \mathbf{q}, \mathbf{q} \perp \mathbf{b}, \mathbf{p} \perp \mathbf{b}$.

B Properties of Subsampled Randomized Hadamard Transform

We first repeat the formal definition of a Hadamard matrix. Then, we provide an auxiliary lemma regarding SRHT, which will be proved useful in our analysis.

Definition 4 (Hadamard Matrix). An Hadamard matrix of size $2n$ is denoted by H_{2n} and is defined via the recursive formula

$$\tilde{H}_{2n} = \begin{pmatrix} \tilde{H}_n & \tilde{H}_n \\ \tilde{H}_n & -\tilde{H}_n \end{pmatrix}, \quad \tilde{H}_1 = (1), \quad H_n = \frac{1}{\sqrt{n}} \tilde{H}_n.$$

Lemma 2. For any $n \geq 1$ and $k \in [n]$, any $S_f \sim \text{SRHT}(k, n)$ satisfies the following properties.

- (1) For all $j_1, j_2 \in [k]$ such that $j_1 \neq j_2$, the j_1 -th row $(S_f)_{j_1, \cdot}$ and the j_2 -th row $(S_f)_{j_2, \cdot}$ are orthogonal.
- (2) For all $i \in [n]$, the i -th column $(S_f)_{\cdot, i}$ satisfies $\|(S_f)_{\cdot, i}\| = 1$.

Proof. We start with establishing (1). To that end, recall that any $S_f \in \text{SRHT}(k, n)$ is given by

$$S_f = \sqrt{\frac{n}{k}} P_{k \times n} H_n B_n$$

where B_n is a diagonal matrix with entries $\{\pm 1\}$, H_n is a normalized Hadamard matrix, and $P_{k \times n}$ is a row subsampling matrix with a single distinct entry equal to 1 in each row. Then, denoting the rows of the matrix H_n by $\{h_i^\top\}_{i=1}^n$ where each $h_i \in \mathbb{R}^{n \times 1}$ we get that the rows of S_f are given by $\left\{\sqrt{\frac{n}{k}}h_{j_i}^\top B_n\right\}_{i=1}^k$ where each $j_i \in [n]$ is a distinct index from $[n]$. Thus, we get that the inner product between two different arbitrary rows with indices j_1 and j_2 such that $j_1 \neq j_2$ is given by

$$\frac{n}{k}h_{j_1}^\top B_n B_n^\top h_{j_2} = \frac{n}{k}h_{j_1}^\top h_{j_2} = 0$$

and where the first equality is since $B_n B_n^\top = \mathbb{I}_n$ by the definition of B_n , and the last equality is since the rows of the Hadamard matrix are orthogonal. This proves point (1). Moreover, for $j_1 = j_2$ we note that this inner product is given by $\frac{n}{k}$.

To prove (2), since H_n is a normalized Hadamard matrix, and since the entries of B_n and $P_{k \times n}$ are either 0's or either ± 1 , the entries of S_f belong to $\left\{\pm \frac{1}{\sqrt{k}}\right\}$. Since each column of S_f contains exactly k elements, the norm of each column is 1. $\square$

C Proof of Theorem 1

Our proof will use the next standard formula for the Rényi divergence between two d -dimensional multivariate Gaussian distributions [Gil et al., 2013]

$$D_\alpha \left(\mathcal{N}(\vec{0}_d, \Sigma_1) \parallel \mathcal{N}(\vec{0}_d, \Sigma_2) \right) = -\frac{1}{2(\alpha - 1)} \log \left(\frac{\det(\Sigma_1 + \alpha(\Sigma_2 - \Sigma_1))}{(\det(\Sigma_1))^{1-\alpha} (\det(\Sigma_2))^\alpha} \right) \quad (6)$$

which holds for all α such that $\alpha \Sigma_1^{-1} + (1 - \alpha) \Sigma_2^{-1} = \Sigma_2^{-1} + \alpha (\Sigma_1^{-1} - \Sigma_2^{-1}) \succ 0$.

We will first provide a proof of Lemma 1. To that end, we start with providing a tight characterization of the privacy guarantee of $\mathcal{M}_{S_f}$, which we then simplify towards the form presented in Lemma 1. Then, we provide some additional auxiliary lemmas, which are necessary for establishing the overall privacy guarantees. For proving the final claim, we will show that the assumptions of Lemma 1 hold in the setting of FastMix, which, under the right choice of parameters, makes its output $(\varepsilon(\omega, \gamma, k_1, \delta), \delta)$ -DP.

C.1 Proof of Lemma 1

Without loss of generality, we consider $X - X' = e_1 x_1^\top$; in simpler terms, X' is obtained by zeroing out the first row in X . We introduce some notation for convenience before delving into the proof. The i^{th} column of S_f is $u_i := S_f e_i$. The matrices Z, Z' are the outputs of X, X' transformed using S_f respectively as

$$Z := S_f X \quad \text{and} \quad Z' := S_f X' = Z - S_f e_1 x_1^\top = Z - u_1 x_1^\top.$$

Finally, we will use the notation $M := (Z^\top Z + \sigma^2 \mathbb{I}_d)^{-1}$ and $\delta_i := Z^\top S_f e_i - x_i$ throughout.

We begin by observing the difference between $Z'^\top Z'$ and $Z^\top Z$ as these play a key role in the Rényi divergence calculation that follows. By definition,

$$Z'^\top Z' = \left(Z - u_1 x_1^\top \right)^\top \left(Z - u_1 x_1^\top \right) = Z^\top Z + \|u_1\|^2 x_1 x_1^\top - Z^\top u_1 x_1^\top - \left(Z^\top u_1 x_1^\top \right)^\top.$$

Define $\tilde{x}_1 = \|u_1\|x_1$ and $\tilde{u}_1 = \frac{u_1}{\|u_1\|}$. Then,

$$Z'^\top Z' = Z^\top Z + \tilde{x}_1 \tilde{x}_1^\top - Z^\top \tilde{u}_1 \tilde{x}_1^\top - (Z^\top \tilde{u}_1 \tilde{x}_1^\top)^\top.$$

Since $\|u_1\| \leq 1$, we have $\|\tilde{x}_1\| = \|u_1\|\|x_1\| \leq 1$ by the preconditions on S_f and X , and $\|\tilde{u}_1\| = 1$.⁴ The remainder of the proof only uses bounds on $\|\tilde{x}_1\|$ and $\|\tilde{u}_1\|$, and therefore without loss of generality, we assume that $\|u_1\| = 1, \|x_1\| \leq 1$ and write

$$Z'^\top Z' = Z^\top Z + x_1 x_1^\top - Z^\top u_1 x_1^\top - (Z^\top u_1 x_1^\top)^\top.$$

Next, we note that the difference between $Z'^\top Z'^\top$ and $Z^\top Z$ can be expressed as a rank-2 update. More precisely, define $U \in \mathbb{R}^{d \times 2} := [-Z^\top u_1 + x_1, x_1]$ and $V \in \mathbb{R}^{d \times 2} := [x_1, -Z^\top u_1]$. Then,

$$UV^\top = -Z^\top u_1 x_1^\top + x_1 x_1^\top - x_1 u_1^\top Z = Z'^\top Z'^\top - Z^\top Z.$$

Since $\mathbf{g}$ and ξ are independent standard normal vectors this implies that

$$\begin{aligned} (\mathbf{g}^\top S_f X + \sigma \xi^\top)^\top &\sim \mathcal{N}\left(0, Z^\top Z + \sigma^2 \mathbb{I}_d\right), \\ (\mathbf{g}^\top S_f X' + \sigma \xi^\top)^\top &\sim \mathcal{N}\left(0, Z'^\top Z' + \sigma^2 \mathbb{I}_d\right) \\ &\stackrel{(d)}{=} \mathcal{N}\left(0, Z^\top Z + \sigma^2 \mathbb{I}_d + x_1 x_1^\top - Z^\top u_1 x_1^\top - (Z^\top u_1 x_1^\top)^\top\right). \end{aligned}$$

Using (6), we have

$$\begin{aligned} D_\alpha \left(\mathbf{g}^\top S_f X + \sigma \xi^\top \middle| \mathbf{g}^\top S_f X' + \sigma \xi^\top \right) \\ = -\frac{1}{2(\alpha - 1)} \log \left(\frac{\det(Z^\top Z + \sigma^2 \mathbb{I}_d + \alpha(Z'^\top Z' - Z^\top Z))}{(\det(Z^\top Z + \sigma^2 \mathbb{I}_d))^{1-\alpha} (\det(Z'^\top Z' + \sigma^2 \mathbb{I}_d))^\alpha} \right). \end{aligned} \quad (7)$$

Now, using the matrix determinant lemma and $\det(AB) = \det(A)\det(B)$ we get for any $\beta \geq 0$

$$\det(Z^\top Z + \sigma^2 \mathbb{I}_d + \beta UV^\top) = \det(Z^\top Z + \sigma^2 \mathbb{I}_d) \det(\mathbb{I}_2 + \beta V^\top MU)$$

where the inner matrix $V^\top MU$ can be rewritten explicitly as

$$V^\top MU = \begin{bmatrix} x_1^\top M(-Z^\top u_1 + x_1) & x_1^\top Mx_1 \\ (u_1)^\top ZM(Z^\top u_1 - x_1) & -(u_1)^\top ZMx_1 \end{bmatrix}.$$

Define

$$s_1 := x_1^\top Mx_1, \quad s_2 := (u_1)^\top ZMZ^\top u_1, \quad c := -x_1^\top MZ^\top u_1 = -(u_1)^\top ZMx_1.$$

Then,

$$\det(\mathbb{I}_2 + \beta V^\top MU) = \det \left(\begin{pmatrix} 1 + \beta s_1 + \beta c & \beta s_1 \\ \beta s_2 + \beta c & 1 + \beta c \end{pmatrix} \right) = 1 + \beta s_1 + 2\beta c + \beta^2 c^2 - \beta^2 s_1 s_2.$$

⁴If $\|u_1\| = 0$, then $Z' = Z$ and the divergence is 0. Thus, we assume that $\|u_1\| > 0$.

Applying this to $\det(Z'^\top Z' + \sigma^2 \mathbb{I}_d)$ with $\beta = 1$, we get

$$\det(Z'^\top Z' + \sigma^2 \mathbb{I}_d) = \det(Z^\top Z + \sigma^2 \mathbb{I}_d) (1 + s_1 + 2c + c^2 - s_1 s_2) ,$$

and applying this with $\beta = \alpha$ for the numerator in (7), we obtain

$$\det(Z^\top Z + \sigma^2 \mathbb{I}_d + \alpha(Z'^\top Z' - Z^\top Z)) = \det(Z^\top Z + \sigma^2 \mathbb{I}_d) (1 + \alpha s_1 + 2\alpha c + \alpha^2 c^2 - \alpha^2 s_1 s_2) .$$

Substituting these inside (7) yields

$$\begin{aligned} D_\alpha \left(\mathbf{g}^\top S_f X + \sigma \xi^\top \parallel \mathbf{g}^\top S_f X' + \sigma \xi^\top \right) \\ = -\frac{1}{2(\alpha - 1)} \log \left(\frac{\det(Z^\top Z + \sigma^2 \mathbb{I}_d + \alpha(Z'^\top Z' - Z^\top Z))}{(\det(Z^\top Z + \sigma^2 \mathbb{I}_d))^{1-\alpha} (\det(Z'^\top Z' + \sigma^2 \mathbb{I}_d))^\alpha} \right) \\ = \frac{1}{2(\alpha - 1)} \log \left(\frac{(1 - (s_1 s_2 - c^2) + (2c + s_1))^\alpha}{1 - \alpha^2 (s_1 s_2 - c^2) + \alpha (2c + s_1)} \right) . \end{aligned} \quad (8)$$

Our goal now is to obtain a closed-form upper bound on (8), which depends only on global properties of $S_f X$ such as the row norm bound $C_{S_f X}^2$ and the minimal eigenvalue $\lambda_{\min}^{S_f X}$. To do this, we obtain bounds on $s_1 s_2 - c^2$ and $2c + s_1$ that solely depend on these quantities, and then obtain an upper bound for (8) by casting it as an optimization problem.

Firstly, $s_1 s_2 - c^2$ is non-negative due to the Cauchy-Schwarz inequality. Then, we further note that the next upper bound

$$\begin{aligned} s_1 s_2 - c^2 &= \left(x_1^\top M x_1 \right) \left(u_1^\top Z M Z^\top u_1 \right) - \left(x_1^\top M Z^\top u_1 \right)^2 \\ &= \left(x_1^\top M x_1 \right) \left((x_1 + \delta_1)^\top M (x_1 + \delta_1) \right) - \left(x_1^\top M (x_1 + \delta_1) \right)^2 \\ &= \left(x_1^\top M x_1 \right) \left(\delta_1^\top M \delta_1 \right) - \left(x_1^\top M \delta_1 \right)^2 \\ &\leq \left(x_1^\top M x_1 \right) \left(\delta_1^\top M \delta_1 \right) \\ &\leq \|x_1\|^2 \|\delta_1\|^2 (\lambda_{\max}(M))^2 \\ &\leq \frac{\|Z^\top S_f e_1 - x_1\|^2}{(\lambda_{\min}^Z + \sigma^2)^2} \\ &\leq \max_{i \in [n]} \frac{\|Z^\top S_f e_i - x_i\|^2}{(\lambda_{\min}^Z + \sigma^2)^2} \end{aligned}$$

holds by the definition of δ_1 and the assumption $\|x_1\|^2 \leq 1$.

Next, we work with $2c + s_1$. Note that $\delta_1 = Z^\top u_1 - x_1$ and

$$2c + s_1 = -2\delta_1^\top M x_1 - 2\|x_1\|_M^2 + \|x_1\|_M^2 = -2\delta_1^\top M x_1 - \|x_1\|_M^2 .$$

Therefore,

$$-\|x_1\|_M^2 - 2\|x_1\|_M \|\delta_1\|_M \leq 2c + s_1 \leq -\|x_1\|_M^2 + 2\|x_1\|_M \|\delta_1\|_M ,$$

and which we can further bound as

$$2c + s_1 \leq 2 \|x_1\|_M \|\delta_1\|_M \leq \frac{2 \|Z^\top S_f e_1 - x_1\|}{\lambda_{\min}^Z + \sigma^2} \leq 2 \cdot \frac{\max_{i \in [n]} \|Z^\top S_f e_i - x_i\|}{\lambda_{\min}^Z + \sigma^2},$$

$$2c + s_1 \geq -\frac{1}{\lambda_{\min}^Z + \sigma^2} - 2 \frac{\|Z^\top S_f e_1 - x_1\|}{\lambda_{\min}^Z + \sigma^2} \geq -\frac{1 + 2 \cdot \max_{i \in [n]} \|Z^\top S_f e_i - x_i\|}{\lambda_{\min}^Z + \sigma^2}.$$

From (8), we obtain natural constraints for the quantities involved:

$$1 - (s_1 s_2 - c^2) + s_1 + 2c > 0, \quad 1 - \alpha^2 (s_1 s_2 - c^2) + \alpha (2c + s_1) > 0.$$

With this, we obtain an upper bound for (8) by posing this as an optimization problem below; this can be viewed as the tightest upper bound possible with the given constraints:

$$\sup_{(s_1, s_2, c) \in \mathcal{D}_1} \left\{ \log \left(\frac{(1 + s_1 + 2c + c^2 - s_1 s_2)^\alpha}{1 + \alpha (s_1 + 2c) + \alpha^2 (c^2 - s_1 s_2)} \right) \right\}, \quad (9)$$

where

$$\mathcal{D}_1 := \left\{ (s_1, s_2, c) \in \mathbb{R}^3 : -\max_{i \in [n]} \frac{\|Z^\top S_f e_i - x_i\|^2}{(\lambda_{\min}^Z + \sigma^2)^2} \leq c^2 - s_1 s_2 \leq 0, \alpha > 1, \right. \\ \left. -\frac{1 + 2 \cdot \max_{i \in [n]} \|Z^\top S_f e_i - x_i\|}{\lambda_{\min}^Z + \sigma^2} \leq s_1 + 2c \leq \frac{2 \cdot \max_{i \in [n]} \|Z^\top S_f e_i - x_i\|}{\lambda_{\min}^Z + \sigma^2}, \right. \\ \left. 1 + s_1 + 2c + c^2 - s_1 s_2 > 0, 1 + \alpha (s_1 + 2c) + \alpha^2 (c^2 - s_1 s_2) > 0 \right\}.$$

From Claim. 1 applied with

$$a \leftarrow s_1, b \leftarrow s_2, c \leftarrow c, \ell \leftarrow \frac{\max_{i \in [n]} \|Z^\top S_f e_i - x_i\|^2}{(\lambda_{\min}^Z + \sigma^2)^2}, m \leftarrow \frac{1 + 2 \cdot \max_{i \in [n]} \|Z^\top S_f e_i - x_i\|}{\lambda_{\min}^Z + \sigma^2}$$

and $M \leftarrow \frac{2 \cdot \max_{i \in [n]} \|Z^\top S_f e_i - x_i\|}{\lambda_{\min}^Z + \sigma^2}$, we have the following bound for the objective in (9) as

$$D_\alpha \left(\mathbf{g}^\top S_f X + \sigma \xi^\top \middle| \mathbf{g}^\top S_f X' + \sigma \xi^\top \right) \\ \leq \frac{1}{2(\alpha - 1)} \log \left(\frac{\left(1 - \max_{i \in [n]} \frac{1 + 2 \|Z^\top S_f e_i - x_i\|}{\lambda_{\min}^Z + \sigma^2} - \max_{i \in [n]} \frac{\|Z^\top S_f e_i - x_i\|^2}{(\lambda_{\min}^Z + \sigma^2)^2} \right)^\alpha}{1 - \alpha \cdot \max_{i \in [n]} \frac{1 + 2 \|Z^\top S_f e_i - x_i\|}{\lambda_{\min}^Z + \sigma^2} - \alpha^2 \cdot \max_{i \in [n]} \frac{\|Z^\top S_f e_i - x_i\|^2}{(\lambda_{\min}^Z + \sigma^2)^2}} \right).$$

Then, since

$$1 + 2 \cdot \max_{i \in [n]} \|Z^\top S_f e_i - x_i\| = C_{S_f X}^2, \quad \max_{i \in [n]} \|Z^\top S_f e_i - x_i\|^2 = \frac{1}{4} (1 - C_{S_f X}^2)^2,$$

and since $\frac{\lambda_{\min}^Z + \sigma^2}{C_{S_f X}^2} = \gamma$ we obtain the upper bound

$$\begin{aligned}
& D_\alpha \left(\mathbf{g}^\top S_f X + \sigma \xi^\top \parallel \mathbf{g}^\top S_f X' + \sigma \xi^\top \right) \\
& \leq \frac{1}{2(\alpha - 1)} \log \left(\frac{\left(1 - \max_{i \in [n]} \frac{1+2\|Z^\top S_f e_i - x_i\|}{\lambda_{\min}^Z + \sigma^2} - \max_{i \in [n]} \frac{\|Z^\top S_f e_i - x_i\|^2}{(\lambda_{\min}^Z + \sigma^2)^2} \right)^\alpha}{1 - \alpha \cdot \max_{i \in [n]} \frac{1+2\|Z^\top S_f e_i - x_i\|}{\lambda_{\min}^Z + \sigma^2} - \alpha^2 \cdot \max_{i \in [n]} \frac{\|Z^\top S_f e_i - x_i\|^2}{(\lambda_{\min}^Z + \sigma^2)^2}} \right) \\
& = \frac{1}{2(\alpha - 1)} \log \left(\frac{\left(1 - \frac{C_{S_f X}^2}{\lambda_{\min}^Z + \sigma^2} - \left(\frac{C_{S_f X}^2}{2(\lambda_{\min}^Z + \sigma^2)} \right)^2 \left(\frac{1 - C_{S_f X}^2}{C_{S_f X}^2} \right)^2 \right)^\alpha}{1 - \frac{\alpha C_{S_f X}^2}{\lambda_{\min}^Z + \sigma^2} - \left(\frac{\alpha C_{S_f X}^2}{2(\lambda_{\min}^Z + \sigma^2)} \right)^2 \left(\frac{1 - C_{S_f X}^2}{C_{S_f X}^2} \right)^2} \right) \\
& = \frac{1}{2(\alpha - 1)} \cdot \left(\alpha \cdot \log \left(1 - \frac{1}{\gamma} - \frac{1}{4\gamma^2} \left(\frac{1 - C_{S_f X}^2}{C_{S_f X}^2} \right)^2 \right) \right. \\
& \quad \left. - \log \left(1 - \frac{\alpha}{\gamma} - \frac{\alpha^2}{4\gamma^2} \left(\frac{1 - C_{S_f X}^2}{C_{S_f X}^2} \right)^2 \right) \right) \\
& := \phi(\alpha; \gamma, C_{S_f X}^2).
\end{aligned}$$

Finally, we place constraints on α, γ for which the Rényi divergence above is well-defined. In particular, we require

$$1 - \frac{1}{\gamma} - \frac{1}{4\gamma^2} \left(\frac{1 - C_{S_f X}^2}{C_{S_f X}^2} \right)^2 > 0 \quad \text{and} \quad 1 - \frac{\alpha}{\gamma} - \frac{\alpha^2}{4\gamma^2} \left(\frac{1 - C_{S_f X}^2}{C_{S_f X}^2} \right)^2 > 0.$$

It can be verified that when

$$\gamma > \bar{\gamma}, \quad 1 < \alpha < \gamma/\bar{\gamma}$$

where $\bar{\gamma} := \frac{1}{2} \left(1 + \sqrt{1 + (C_{S_f X}^2 - 1)^2} \right)$, this holds.

It remains to analyze the case where one row of X' is zeroed out relative to X . In this case, $S_f X' = S_f X + S_f e_i x_i^\top = S_f X + u_i x_i^\top$. The analysis is unchanged, except for a sign change in u_i , which yields a sign change in c . By symmetry, it leads to the same final result. Since the bound is independent of the row removed from X , it applies whenever any point $i \in [n]$ is removed. This completes the proof of the closed-form and tightest guarantee.

Simplifying the Bound. Note that the previous proof established the upper bound

$$\max_{X' \simeq X} D_\alpha(\mathcal{M}_{S_f}(X) \parallel \mathcal{M}_{S_f}(X')) \leq \phi(\alpha; \gamma, C_{S_f X}^2)$$

for all $\alpha \in (1, \gamma/\bar{\gamma})$ provided that $\gamma > \bar{\gamma}$. First, since $C_{S_f X}^2 \geq 1$ and $\left(\frac{1 - C_{S_f X}^2}{C_{S_f X}^2} \right)^2 \leq 1$ for all $C_{S_f X}^2 \geq 1$, the previous monotonicity arguments further yields the bound

$$\max_{(X, X'): X \simeq X'} D_\alpha(\mathcal{M}_{S_f}(X) \parallel \mathcal{M}_{S_f}(X'))$$

$$\begin{aligned} &\leq \frac{1}{2(\alpha-1)} \left(\alpha \cdot \log \left(1 - \frac{1}{\gamma} - \frac{1}{4\gamma^2} \right) - \log \left(1 - \frac{\alpha}{\gamma} - \frac{\alpha^2}{4\gamma^2} \right) \right) \\ &:= \bar{\phi}(\alpha; \gamma) \end{aligned}$$

for all $\alpha \in (1, 4\gamma/5)$ provided that $\gamma > 5/4$ and where these constraints were obtained since $\bar{\gamma} \leq 5/4$. Now, note that by Calim. 2, over this range for α and γ it further holds that

$$\bar{\phi}(\alpha; \gamma) \leq \frac{1}{2(\alpha-1)} \left(\alpha \cdot \log \left(1 - \frac{5}{4\gamma} \right) - \log \left(1 - \frac{5\alpha}{4\gamma} \right) \right).$$

This upper bound matches the function $\varphi(\alpha; 4\gamma/5)$ defined in Proposition 2. Thus, using Lev et al. [2025, Corollary. 1], we know that this is upper bounded by

$$\frac{1}{2(\alpha-1)} \left(\alpha \cdot \log \left(1 - \frac{5}{4\gamma} \right) - \log \left(1 - \frac{5\alpha}{4\gamma} \right) \right) \leq \frac{25\alpha}{32\gamma^2}$$

for all $\alpha \in (1, 8\gamma/25)$ and provided that $\gamma \geq 25/8$. This establishes the bound presented in Lemma 1. $\square$

Remark 3. The same proof recovers the privacy analysis from Lev et al. [2025] whenever $S_f^\top S_f = \mathbb{I}_n$. In this case, we get $C_{S_f X}^2 = 1$, and the overall divergence becomes

$$\frac{1}{2(\alpha-1)} \left(\alpha \cdot \log \left(1 - \frac{1}{\lambda_{\min}^X + \sigma^2} \right) - \log \left(1 - \frac{\alpha}{\lambda_{\min}^X + \sigma^2} \right) \right)$$

and for $\lambda_{\min}^X + \sigma^2 > 1$ and $1 < \alpha < \lambda_{\min}^X + \sigma^2$, which recovers exactly the analysis from Lev et al. [2025, Appendix. B]. We note that in this case, S_f is an orthogonal matrix, so $\mathbf{g}^\top S_f \sim \mathcal{N}(\vec{0}_n, \mathbb{I}_n)$, which is distributionally equivalent to a standard normal vector, so the overall mechanism is procedurally the same as the Gaussian mixing mechanism from Lev et al. [2025].

C.1.1 Supplementary Algebraic Claims

Here, we state and prove algebraic claims that we have used in the proof of Theorem 1.

Claim 1. *Let $\alpha > 1$. Suppose $m, \ell > 0$ and M is a general constant satisfying $|M| \leq m$. Consider the set $\mathcal{D}_1 \subseteq \mathbb{R}^3$*

$$\mathcal{D}_1 = \left\{ (a, b, c) \in \mathbb{R}^3 : \begin{array}{l} -\ell \leq c^2 - ab \leq 0 \\ -m \leq a + 2c \leq M \\ 1 + a + 2c + c^2 - ab > 0 \\ 1 + \alpha(a + 2c) + \alpha^2(c^2 - ab) > 0 \end{array} \right\}.$$

Then,

$$\sup_{(a,b,c) \in \mathcal{D}_1} \log \left(\frac{(1 + a + 2c + c^2 - ab)^\alpha}{1 + \alpha(a + 2c) + \alpha^2(c^2 - ab)} \right) \leq \log \left(\frac{(1 - m - \ell)^\alpha}{1 - \alpha m - \alpha^2 \ell} \right).$$

Proof. For convenience, we use the following notation. Let $x = c^2 - ab$, $y = 2c + a$, $A = 1 + x + y$, and $B = 1 + \alpha y + \alpha^2 x$. The objective can be expressed as $\alpha \cdot \log(A) - \log(B)$. For any y ,

$$\frac{\partial}{\partial x} \{ \alpha \cdot \log(A) - \log(B) \} = \frac{\alpha}{A} - \frac{\alpha^2}{B} = \frac{\alpha}{AB} (B - \alpha A) = \frac{\alpha(\alpha - 1)}{AB} (-1 + \alpha x).$$

Since $\alpha > 1$ and $x = c^2 - ab \leq 0$ according to the constraints, this partial derivative is non-positive. As a result, we can eliminate x from the maximization to obtain the upper bound characterized by the problem

$$\sup_{y \in \mathcal{D}_2} \log \left(\frac{(1 + y - \ell)^\alpha}{1 + \alpha y - \alpha^2 \ell} \right)$$

where $\mathcal{D}_2 := \{y \in \mathbb{R} : -m \leq y \leq M, 1 + y - \ell > 0, 1 + \alpha y - \alpha^2 \ell > 0\}$.

We take the derivative of this objective in y to obtain

$$\frac{\partial}{\partial y} \log \left(\frac{(1 + y - \ell)^\alpha}{1 + \alpha y - \alpha^2 \ell} \right) = \frac{\alpha(\alpha - 1)(y - (\alpha + 1)\ell)}{(1 + y - \ell)(1 + \alpha y - \alpha^2 \ell)}. \quad (10)$$

We note that this derivative attains a single 0 at $y^* = (\alpha + 1)\ell$, and moreover is negative for all $y < y^*$ and positive for all $y > y^*$. Therefore, since the objective is continuous and differentiable, y^* is the unique minimizer whenever it lies in the feasible domain; otherwise, the objective is monotone on the feasible interval. In either case, the supremum over the bounded feasible interval is attained on its boundary.

We note that the interval boundaries in our case are given by $-m$ and M . We demonstrate below that the maximum is attained at $-m$. To see this, let

$$\Delta(y) := \log \left(\frac{(1 - y - \ell)^\alpha}{1 - \alpha y - \alpha^2 \ell} \right) - \log \left(\frac{(1 + y - \ell)^\alpha}{1 + \alpha y - \alpha^2 \ell} \right), \quad y \geq 0.$$

Then, using (10), we note that

$$\begin{aligned} \frac{\partial \Delta(y)}{\partial y} &= \frac{\alpha(\alpha - 1)(y + (\alpha + 1)\ell)}{(1 - y - \ell)(1 - \alpha y - \alpha^2 \ell)} - \frac{\alpha(\alpha - 1)(y - (\alpha + 1)\ell)}{(1 + y - \ell)(1 + \alpha y - \alpha^2 \ell)} \\ &= \frac{2\alpha(\alpha - 1)(\alpha + 1)(y^2 + \ell(1 - \ell)(1 - \alpha^2 \ell))}{(1 - y - \ell)(1 + y - \ell)(1 - \alpha y - \alpha^2 \ell)(1 + \alpha y - \alpha^2 \ell)}. \end{aligned}$$

Now, since the denominator is positive for all y in the interval (since the positiveness of the different factors is required for the positiveness of the factors inside the log), and since $\alpha > 1$, we note that the sign of this derivative is determined by the sign of $y^2 + \ell(1 - \ell)(1 - \alpha^2 \ell)$. However, since the definiteness of the log at $y = 0$ requires $1 - \alpha^2 \ell \geq 0$ and $1 - \ell \geq 0$ and since $\ell \geq 0$ we get that $\ell(1 - \ell)(1 - \alpha^2 \ell) \geq 0$, so $y^2 + \ell(1 - \ell)(1 - \alpha^2 \ell) \geq 0$, and this derivative is non-negative. Since $\Delta(0) = 0$, we get that $\Delta \geq 0$, so the negative branch is larger. Since $|M| \leq m$, this together with the previous analysis of the derivative further says that the maximum of the overall objective is attained on $-m$. This completes the proof. $\square$

Claim 2. *Let $x > 5/4$. Then, for all $\alpha \in (1, 4x/5)$,*

$$\alpha \cdot \log \left(1 - \frac{1}{x} - \frac{1}{4x^2} \right) - \log \left(1 - \frac{\alpha}{x} - \frac{\alpha^2}{4x^2} \right) \leq \alpha \cdot \log \left(1 - \frac{5}{4x} \right) - \log \left(1 - \frac{5\alpha}{4x} \right).$$

Proof. Define $g(z) := \log \left(1 - z - \frac{z^2}{4} \right) - \log \left(1 - \frac{5z}{4} \right)$ for $z \in [0, 4/5)$. We note that the desired inequality is equivalent to

$$\alpha \cdot g(1/x) \leq g(\alpha/x).$$

Since $g(0) = 0$, the claim follows if $g(z)$ is convex over the range $z \in [0, 4/5)$, since convexity then implies that for all $\alpha > 1$

$$g(z) = g\left(\frac{1}{\alpha}(\alpha z) + \left(1 - \frac{1}{\alpha}\right) \cdot 0\right) \geq \frac{1}{\alpha} \cdot g(\alpha z) + \left(1 - \frac{1}{\alpha}\right) g(0)$$

so $\alpha \cdot g(z) \geq g(\alpha z)$, and setting $z = 1/x$ would prove the statement of the lemma.

The remainder of the proof focuses on the convexity of g . The second derivative of g is

$$g''(z) = -\frac{2(4 - 4z - z^2) + (4 + 2z)^2}{(4 - 4z - z^2)^2} + \frac{25}{(4 - 5z)^2} = \frac{16 + 32z - 112z^2 + 80z^3 - 25z^4}{(4 - 5z)^2(4 - 4z - z^2)^2}.$$

The denominator is positive for $z \in [0, 4/5)$. The numerator has two real roots over $\mathbb{R}$ at $z_1 \approx 0.814$ and $z_2 \approx -0.247$. As a result, since on $z = 0$ its positive, the quartic polynomial in the numerator is also non-negative in the range $z \in [0, 4/5)$, which establishes that $g''(z) \geq 0$ in $z \in [0, 4/5)$. This asserts that g is convex, thus proving the claim. $\square$

C.2 Bound For $C_{S_f X}^2 \rightarrow \infty$

We note that the previous bound requires that the quantity $\max_{i \in [n]} \|Z^\top S_f e_i - x_i\|$ will be finite.

However, as we show now, a finite divergence is also possible in the case where we do not rely on such a bound. This requires a particular analysis since taking $C_{S_f X}^2 \rightarrow \infty$ in the previous analysis yields $\gamma \rightarrow \infty$. However, as we show now, a more fine-grained analysis in this case leads to a finite divergence that depends on $\lambda_{\min}^Z + \sigma^2$.

In particular, note that the next trivial bounds can be obtained on s_1, s_2 , and c :

$$s_1 \leq \frac{\|x_1\|^2}{\lambda_{\min}^Z + \sigma^2} \leq \frac{1}{\lambda_{\min}^Z + \sigma^2}, \quad s_2 \leq \frac{\lambda_{\max}(Z^\top Z)}{\lambda_{\max}(Z^\top Z) + \sigma^2} \|u_1\|^2 \leq \|u_1\|^2 \leq 1, \quad c \leq \sqrt{s_1 s_2} \leq (\lambda_{\min}^Z + \sigma^2)^{-1/2}.$$

Then, using the notation

$$\log\left(\frac{(1 + s_1 + 2c + c^2 - s_1 s_2)^\alpha}{1 + \alpha(s_1 + 2c) + \alpha^2(c^2 - s_1 s_2)}\right) := \log\left(\frac{A^\alpha}{B}\right) = \alpha \cdot \log(A) - \log(B),$$

we note that the divergence is a monotonically increasing function of s_2 , since

$$\frac{\partial}{\partial s_2} \{\alpha \cdot \log(A) - \log(B)\} = -\frac{\alpha s_1}{A} + \frac{\alpha^2 s_1}{B} = \frac{\alpha s_1}{AB} (\alpha A - B) = \frac{\alpha(\alpha - 1)s_1}{AB} (1 + \alpha(s_1 s_2 - c^2)) \geq 0.$$

Thus, we can substitute $s_2 = 1$ and to get the upper bound

$$\log\left(\frac{(1 + s_1 + 2c + c^2 - s_1 s_2)^\alpha}{1 + \alpha(s_1 + 2c) + \alpha^2(c^2 - s_1 s_2)}\right) \leq 2\alpha \cdot \log(1 + c) - \log\left((1 + \alpha c)^2 + \alpha(1 - \alpha)s_1\right).$$

Now, since $1 - \alpha \leq 0$, this function is monotonically increasing in s_1 . Thus, we substitute $s_1 = (\lambda_{\min}^Z + \sigma^2)^{-1}$ and obtain the upper bound

$$\log\left(\frac{(1 + s_1 + 2c + c^2 - s_1 s_2)^\alpha}{1 + \alpha(s_1 + 2c) + \alpha^2(c^2 - s_1 s_2)}\right) \leq 2\alpha \cdot \log(1 + c) - \log\left((1 + \alpha c)^2 + \frac{\alpha - \alpha^2}{\lambda_{\min}^Z + \sigma^2}\right).$$

Now, this bound can be optimized over c under the constraint in which $|c| \leq (\lambda_{\min}^Z + \sigma^2)^{-1/2}$. However, by similar arguments as used before, we can prove that the maximum is attained on $c = -(\lambda_{\min}^Z + \sigma^2)^{-1/2}$, which yields the closed-form bound

$$\begin{aligned} D_\alpha \left(\mathbf{g}^\top S_f X + \sigma \xi^\top \middle| \middle| \mathbf{g}^\top S_f X' + \sigma \xi^\top \right) \\ \leq \frac{1}{2(\alpha-1)} \cdot \left(2\alpha \cdot \log \left(1 - \frac{1}{\sqrt{\lambda_{\min}^Z + \sigma^2}} \right) - \log \left(\left(1 - \frac{\alpha}{\sqrt{\lambda_{\min}^Z + \sigma^2}} \right)^2 + \frac{\alpha - \alpha^2}{\lambda_{\min}^Z + \sigma^2} \right) \right) \\ = \frac{1}{2(\alpha-1)} \cdot \left(2\alpha \cdot \log \left(1 - \frac{1}{\sqrt{\lambda_{\min}^Z + \sigma^2}} \right) - \log \left(1 - \frac{2\alpha}{\sqrt{\lambda_{\min}^Z + \sigma^2}} + \frac{\alpha}{\lambda_{\min}^Z + \sigma^2} \right) \right) \end{aligned}$$

and which is defined for all $1 < \alpha < \frac{\lambda_{\min}^Z + \sigma^2}{2\sqrt{\lambda_{\min}^Z + \sigma^2} - 1}$ provided that $\lambda_{\min}^Z + \sigma^2 > 1$.

C.3 Auxiliary Lemmas

We now provide three additional auxiliary lemmas, which will prove useful for making our overall privacy claim of Theorem 1. Throughout, we use the notion of ℓ_1 sensitivity, which, given a target function $f(X)$, is defined by

$$\max_{(X, X') : X' \simeq X} |f(X) - f(X')|.$$

Lemma 3. *Let $u, v \in \mathbb{R}^d$. Then,*

$$\left\| uv^\top \right\|_{\text{op}} = \|u\| \|v\|, \quad \left\| uv^\top + vu^\top \right\|_{\text{op}} = \|u\| \|v\| + |u^\top v|.$$

Proof. If either $u = \vec{0}_d$ or $v = \vec{0}_d$, both identities are immediate. Hence, assume $u, v \neq \vec{0}_d$, and define

$$\bar{u} := \frac{u}{\|u\|}, \quad \bar{v} := \frac{v}{\|v\|}.$$

We first calculate the operator norm of $uv^\top$. We have

$$uv^\top = \|u\| \|v\| \bar{u} \bar{v}^\top.$$

Since $\bar{u} \bar{v}^\top$ is a rank-one matrix with its only nonzero singular value equal to one, it follows that

$$\left\| uv^\top \right\|_{\text{op}} = \|u\| \|v\|.$$

Next, consider

$$A := uv^\top + vu^\top = \|u\| \|v\| \left(\bar{u} \bar{v}^\top + \bar{v} \bar{u}^\top \right).$$

Then,

$$\|A\|_{\text{op}} = \max_{\|x\|=1} |x^\top A x| = 2 \|u\| \|v\| \max_{\|x\|=1} |(\bar{u}^\top x)(\bar{v}^\top x)|.$$

Thus, it remains to show that

$$\max_{\|x\|=1} \left| (\bar{u}^\top x)(\bar{v}^\top x) \right| = \frac{1 + |\bar{u}^\top \bar{v}|}{2}.$$

Let $\rho := |\bar{u}^\top \bar{v}|$. For any $\|x\| = 1$, by $2|ab| \leq a^2 + b^2$,

$$\begin{aligned} 2 \left| (\bar{u}^\top x)(\bar{v}^\top x) \right| &\leq (\bar{u}^\top x)^2 + (\bar{v}^\top x)^2 \\ &= x^\top \left(\bar{u} \bar{u}^\top + \bar{v} \bar{v}^\top \right) x \\ &\leq \lambda_{\max} \left(\bar{u} \bar{u}^\top + \bar{v} \bar{v}^\top \right). \end{aligned}$$

The nonzero eigenvalues of $\bar{u} \bar{u}^\top + \bar{v} \bar{v}^\top$ are $1 + \rho$ and $1 - \rho$, which follows since the eigenvectors of this matrix are $\frac{\bar{u} + \bar{v}}{\|\bar{u} + \bar{v}\|}$ and $\frac{\bar{u} - \bar{v}}{\|\bar{u} - \bar{v}\|}$. Hence,

$$\max_{\|x\|=1} \left| (\bar{u}^\top x)(\bar{v}^\top x) \right| = \frac{1 + \rho}{2}.$$

Substituting this into the previous equation gives

$$\begin{aligned} \left\| uv^\top + vu^\top \right\|_{\text{op}} &= 2 \|u\| \|v\| \cdot \frac{1 + |\bar{u}^\top \bar{v}|}{2} \\ &= \|u\| \|v\| + \left| u^\top v \right|. \end{aligned}$$

This completes the proof. $\square$

Lemma 4. *Let X be a dataset such that $\|x_i\| \leq 1$ for all $i \in [n]$. Suppose $S_f \in \mathbb{R}^{k \times n}$ is a matrix such that $\|S_f e_i\| \leq 1$ for all $i \in [n]$. Then, the ℓ_1 sensitivity of $\max_{i \in [n]} \|X^\top S_f^\top S_f e_i - x_i\|$ is bounded from above by $\hat{\Delta} := \max_{i,j} |e_i^\top (S_f^\top S_f - \mathbb{I}_n) e_j|$.*

Proof. Choose an arbitrary $j \in [n]$ and define $P := S_f^\top S_f - \mathbb{I}_n$. For each $i \in [n]$, set

$$v_i := X^\top P e_i, \quad v'_i := \left(X - e_j x_j^\top \right)^\top P e_i = v_i - x_j \left(e_j^\top P e_i \right).$$

We note that v'_i is the quantity $f_i(X) := X^\top (S_f^\top S_f - \mathbb{I}_n) e_i$, evaluated on the neighboring dataset X' obtained by zeroing out the j^{th} row in X . By the triangle and Cauchy-Schwarz inequalities, we get

$$\left| \|v'_i\| - \|v_i\| \right| \leq \left| \|v'_i\| - \|v_i\| \right| \leq \|v'_i - v_i\| = \|x_j\| \left| e_j^\top P e_i \right|.$$

Substituting the definition of P and the condition on $\|x_i\|$,

$$\left| \|v'_i\| - \|v_i\| \right| \leq \max_{i,j} \left| e_i^\top \left(S_f^\top S_f - \mathbb{I}_n \right) e_j \right|$$

for every $i \in [n]$. Now, let $a_i := \|v'_i\|$ and $b_i := \|v_i\|$. Then,

$$\left| \max_{i \in [n]} a_i - \max_{i \in [n]} b_i \right| \leq \max_{i \in [n]} |a_i - b_i| \leq \max_{i,j} \left| e_i^\top \left(S_f^\top S_f - \mathbb{I}_n \right) e_j \right|.$$

Since j is arbitrary, and since the bound is independent of the particular choice of X , it remains valid after taking the maximum over all $j \in [n]$ and all neighboring pairs (X, X') . Hence,

$$\max_{(X, X') : X' \simeq X} \left| \max_{i \in [n]} f_i(X) - \max_{i \in [n]} f_i(X') \right| \leq \max_{i, j} \left| e_i^\top \left(S_f^\top S_f - \mathbb{I}_n \right) e_j \right|.$$

This is precisely the desired ℓ_1 -sensitivity. $\square$

Lemma 5. *Let $S_f \in \mathbb{R}^{k \times n}$ satisfying $\|S_f e_i\| \leq 1$ for all $i \in [n]$. Let $X \in \mathbb{R}^{n \times d}$ satisfying $\|x_i\| \leq 1$ for all $i \in [n]$ and define $Z := S_f X$. Set $m := \max_{i \in [n]} \|Z^\top S_f e_i - x_i\|$ and $\Lambda := 2 - \min_{i \in [n]} \|S_f e_i\|^2$. Then, the ℓ_1 sensitivity of $\lambda_{\min}^Z$ is upper bounded by $\Lambda + 2m$.*

Proof. Denoting $X^\top S_f^\top S_f e_i = x_i + \delta_i$ and recalling that $\|S_f e_i\|^2 \leq 1$ for all $i \in [n]$ we note that

$$\begin{aligned} \left(X - e_i x_i^\top \right)^\top S_f^\top S_f \left(X - e_i x_i^\top \right) &= X^\top S_f^\top S_f X + \|S_f e_i\|^2 x_i x_i^\top - X^\top S_f^\top S_f e_i x_i^\top - x_i e_i^\top S_f^\top S_f X \\ &= X^\top S_f^\top S_f X + \|S_f e_i\|^2 x_i x_i^\top - (x_i + \delta_i) x_i^\top - x_i (x_i + \delta_i)^\top \\ &= X^\top S_f^\top S_f X + \left(\|S_f e_i\|^2 - 2 \right) x_i x_i^\top - \delta_i x_i^\top - x_i \delta_i^\top. \end{aligned}$$

Now, by Weyl's inequality [Stewart, 1991, Theorem. 1] for matrices $X^\top X$ and $Y^\top Y$ it holds

$$|\lambda_{\min}^X - \lambda_{\min}^Y| \leq \|X^\top X - Y^\top Y\|_{\text{op}}.$$

Thus,

$$\begin{aligned} \left| \lambda_{\min}^{Z - S_f e_i x_i^\top} - \lambda_{\min}^Z \right| &\leq \left\| \left(\|S_f e_i\|^2 - 2 \right) x_i x_i^\top - \delta_i x_i^\top - x_i \delta_i^\top \right\|_{\text{op}} \\ &\leq \left| \|S_f e_i\|^2 - 2 \right| \|x_i x_i^\top\|_{\text{op}} + \left\| \delta_i x_i^\top + x_i \delta_i^\top \right\|_{\text{op}} \\ &= \left| \|S_f e_i\|^2 - 2 \right| \|x_i\|^2 + \|\delta_i\| \|x_i\| + \left| \delta_i^\top x_i \right| \\ &\leq \left| \|S_f e_i\|^2 - 2 \right| \|x_i\|^2 + 2 \|\delta_i\| \|x_i\| \\ &\leq \max_{i \in [n]} \left| \|S_f e_i\|^2 - 2 \right| + 2m \\ &\leq \left(2 - \min_{i \in [n]} \|S_f e_i\|^2 \right) + 2m \\ &= \Lambda + 2m \end{aligned}$$

where we have used the triangle inequality, Lemma 3, the assumptions $\|x_i\| \leq 1$ and $\|S_f e_i\| \leq 1$ for all $i \in [n]$, and the definitions of m and Λ . $\square$

Lemma 6. Given a dataset $\mathcal{D}$ and a binary random variable $\mathbf{c} \in \{0, 1\}$, let $\mathcal{M}(\mathcal{D})$ defined as

$$\mathcal{M}(\mathcal{D}) := \begin{cases} \mathcal{M}_{\text{good}}(\mathcal{D}) & \text{if } \mathbf{c} = 0 \\ \mathcal{M}_{\text{bad}}(\mathcal{D}) & \text{else} \end{cases}$$

where $\mathbb{P}(\mathbf{c} = 1) = 1 - \mathbb{P}(\mathbf{c} = 0) = p$. If the mechanism $\mathcal{M}_{\text{good}}(\mathcal{D})| \{\mathbf{c} = 0\}$ is (ε, δ) -DP then $\mathcal{M}(\mathcal{D})$ is $(\varepsilon, \delta + p)$ -DP .

Proof. The proof holds following the definition of (ε, δ) -DP. In particular, for any neighbors $\mathcal{D}, \mathcal{D}'$ and $\mathcal{S} \subseteq \text{range}(\mathcal{M})$, since $\mathcal{M}_{\text{good}}$ is (ε, δ) -DP, and by using the law of total probability, it holds

$$\begin{aligned} \mathbb{P}(\mathcal{M}(\mathcal{D}) \in \mathcal{S}) &= \mathbb{P}(\mathbf{c} = 0) \mathbb{P}(\mathcal{M}(\mathcal{D}) \in \mathcal{S} | \mathbf{c} = 0) + \mathbb{P}(\mathbf{c} = 1) \mathbb{P}(\mathcal{M}(\mathcal{D}) \in \mathcal{S} | \mathbf{c} = 1) \\ &= (1 - p) \cdot \mathbb{P}(\mathcal{M}_{\text{good}}(\mathcal{D}) \in \mathcal{S} | \mathbf{c} = 0) + p \cdot \mathbb{P}(\mathcal{M}_{\text{bad}}(\mathcal{D}) \in \mathcal{S} | \mathbf{c} = 1) \\ &\leq \mathbb{P}(\mathcal{M}_{\text{good}}(\mathcal{D}) \in \mathcal{S} | \mathbf{c} = 0) + p \\ &\leq e^\varepsilon \cdot \mathbb{P}(\mathcal{M}_{\text{good}}(\mathcal{D}') \in \mathcal{S} | \mathbf{c} = 0) + \delta + p \end{aligned}$$

which matches the $(\varepsilon, \delta + p)$ -DP definition. $\square$

C.4 Proof of Theorem 1

C.4.1 Differential Privacy

We recall certain definitions here for the convenience of the reader.

$$\begin{array}{l|l} \hat{\Lambda} & 2 - \min_{i \in [n]} \|S_{\text{f}} e_i\|^2 \\ \hat{\Delta} & \max_{i, j \in [n]} |e_i^\top (S_{\text{f}}^\top S_{\text{f}} - \mathbb{I}_n) e_j| \\ \hat{m} & \max_{i \in [n]} \|X^\top S_{\text{f}}^\top S_{\text{f}} e_i - x_i\| \\ \tilde{m} & \max \left\{ 1, \hat{m} + \omega \hat{\Delta}(\tau - z_1) \right\} \text{ where } z_1 \sim \text{Lap}(0, 1) \\ \tilde{\lambda} & \max \left\{ 0, \lambda_{\min}^{S_{\text{f}}^\top X} - \omega(\hat{\Lambda} + 2\tilde{m})(\tau - z_2) \right\} \text{ where } z_2 \sim \text{Lap}(0, 1) \end{array}$$

Let $\mathcal{E} := \left\{ (\tilde{\lambda} \leq \hat{\lambda}) \cap (\tilde{m} \geq \hat{m}) \right\}$. First, note that for the τ stated in the theorem and by using a union-bound, it holds

$$\mathbb{P}(\mathcal{E}^c) \leq \mathbb{P}(\hat{\lambda} \leq \tilde{\lambda}) + \mathbb{P}(\tilde{m} \leq \hat{m}) = \mathbb{P}(z_1 \geq \tau) + \mathbb{P}(z_2 \geq \tau) \leq \frac{2\delta}{3}.$$

From now on, we prove the privacy guarantees of the mechanism conditioned on the event $\mathcal{E}$, where by Lemma 6 this incurs a loss of $2\delta/3$ in the overall δ .

Privact Guarantees for the Releases $\tilde{m}$ and $\tilde{\lambda}$. Since by Lemma 4 the ℓ_1 sensitivity of $\hat{m}$ is upper bounded by $\hat{\Delta}$, $\tilde{m}$ is $\frac{1}{\omega}$ -DP release of $\hat{m}$ by using classical properties of the Laplace mechanism [Dwork et al., 2014, Section. 3.3]. Moreover, since by Lemma 5 the ℓ_1 sensitivity of $\hat{\lambda}$ is $\hat{\Lambda} + 2\hat{m}$, which under $\mathcal{E}$ is upper bounded by $\hat{\Lambda} + 2\tilde{m}$, using the same properties of the Laplace mechanism we get that $\tilde{\lambda}$ is $\frac{1}{\omega}$ -DP release of $\hat{\lambda}$ (see Vadhan [2017, Section. 3] why using a global upper bound on the local sensitivity provides DP). Thus, these releases contribute another $2/\omega$ to the overall ε .

Guarantees of Sketching Steps. We note that the S_f considered in the theorem satisfies the unit column-norm condition required by Lemma 1, and moreover (2) is satisfied under $\mathcal{E}$, since under $\mathcal{E}$ it holds $\tilde{\eta}^2 \geq \gamma(1+2\hat{m}) - \hat{\lambda}$. Thus, the privacy guarantee of Lemma 1 holds with respect to every row in the output $S_G S_f X + \tilde{\eta}\xi$. Then, since the output comprised of k_1 independent rows of the Gaussian sketch applied over $S_f X$, using composition for the k_1 different rows we get that the output satisfies $(\alpha, k_1 \cdot \bar{\phi}(\alpha; \gamma))$ -RDP for $1 < \alpha < 4\gamma/5$, where we have used the upper bound $\phi(\alpha; \gamma, C_{S_f X}^2) \leq \bar{\phi}(\alpha; \gamma)$. Using the conversion from RDP to (ε, δ) -DP of Proposition 1 this contributes another $\delta/3$ to the overall δ and

$$\min_{1 < \alpha < 4\gamma/5} \left\{ k_1 \bar{\phi}(\alpha; \gamma) + \frac{\log(3/\delta) + (\alpha - 1) \log(1 - 1/\alpha) - \log(\alpha)}{\alpha - 1} \right\}$$

to the overall ε .

Composition. Composing all the different components (while by using Lemma 6 we add $\mathbb{P}(\mathcal{E}^c)$ to the overall δ) yields that the output satisfies $(\hat{\varepsilon}(\omega, \gamma, k_1, \delta), \delta)$ -DP, where

$$\begin{aligned} \hat{\varepsilon}(\omega, \gamma, k_1, \delta) &:= \frac{2}{\omega} + \min_{1 < \alpha < 4\gamma/5} \left\{ k_1 \bar{\phi}(\alpha; \gamma) + \frac{\log(3/\delta) + (\alpha - 1) \log(1 - 1/\alpha) - \log(\alpha)}{\alpha - 1} \right\}, \\ \bar{\phi}(\alpha; \gamma) &:= \frac{1}{2(\alpha - 1)} \cdot \left(\alpha \cdot \log \left(1 - \frac{1}{\gamma} - \frac{1}{4\gamma^2} \right) - \log \left(1 - \frac{\alpha}{\gamma} - \frac{\alpha^2}{4\gamma^2} \right) \right) \end{aligned}$$

and which holds for all $\gamma > 5/4$. Substituting the upper bound on $\bar{\phi}$ derived in Appendix C.1 yields

$$\begin{aligned} \varepsilon(\omega, \gamma, k_1, \delta) &:= \frac{2}{\omega} + \min_{1 < \alpha < 8\gamma/25} \left\{ \frac{25\alpha k_1}{32\gamma^2} + \frac{\log(3/\delta) + (\alpha - 1) \log(1 - 1/\alpha) - \log(\alpha)}{\alpha - 1} \right\} \\ &\leq \frac{2}{\omega} + \min_{1 < \alpha < 8\gamma/25} \left\{ \frac{25\alpha k_1}{32\gamma^2} + \frac{\log(3/\delta)}{\alpha - 1} \right\}. \end{aligned}$$

Substituting $\alpha = \min \left\{ \frac{8\gamma}{25}, 1 + \gamma \sqrt{\frac{32 \log(3/\delta)}{25k_1}} \right\}$ yields the upper bound of Theorem 1.

C.4.2 Runtime Complexity

We provide the proof by explicitly writing the complexity of the different steps inside **FastMix** and their runtime complexity.

Computing Z . Involves applying an SRHT of size (k_2, n) to $X \in \mathbb{R}^{n \times d}$. Following classical arguments (see, for example, Ailon and Liberty [2009]), the runtime complexity of this step is $O(nd \log(k_2))$. However, in our implementation, we first form the full matrix $H_n B_n X$ and then subsample its rows according to the non-zero locations in the matrix $P_{k \times n}$, since we use the values computed for $H_n B_n X$ in our subsequent steps. Thus, the complexity of this step is $O(nd \log(n))$.

Computing $\tilde{\lambda}$. Since $Z \in \mathbb{R}^{k_2 \times d}$, one can compute $Z^\top Z$ in $O(k_2 d^2)$ time and then compute its minimal eigenvalue in $O(d^3)$ (via standard eigensolvers). Thus, the total complexity is $O(k_2 d^2 + d^3)$. In the regime $k_2 \geq d$ (which is necessary for ensuring that $\lambda_{\min}^X \approx \lambda_{\min}^Z$), this simplifies to $O(k_2 d^2)$. We note that in this regime, this can be achieved via a single SVD of Z , which costs $O(k_2 d^2)$.

Computing $\hat{m}$ and $\hat{\Delta}$. For computing $\hat{m}$, we first note that

$$Z^\top S_f e_i - x_i = X^\top \left(\frac{n}{k} B_n^\top H_n^\top P_{k \times n}^\top P_{k \times n} H_n B_n - \mathbb{I}_n \right) e_i = (H_n B_n X)^\top \left(\frac{n}{k} P_{k \times n}^\top P_{k \times n} - \mathbb{I}_n \right) H_n B_n e_i.$$

Then, since $P_{k \times n}$ selects k rows, the matrix $\frac{n}{k} P_{k \times n}^\top P_{k \times n} - \mathbb{I}_n$ is diagonal, with exactly k nonzero diagonal corrections relative to $-\mathbb{I}_n$. Hence, multiplying by this matrix amounts only to rescaling selected rows (equivalently, columns after transposition), so forming $(H_n B_n X)^\top \left(\frac{n}{k} P_{k \times n}^\top P_{k \times n} - \mathbb{I}_n \right)$ costs $O(nd)$. Then, since we have $H_n B_n X$ from the previous computations, the overall computations requires applying the Hadamard transform $H_n B_n$ on the matrix $(H_n B_n X)^\top \left(\frac{n}{k} P_{k \times n}^\top P_{k \times n} - \mathbb{I}_n \right)$, which costs $O(nd \log(n))$, and then computing the norms of the columns of this matrix which takes another $O(nd)$. Thus, the overall complexity of this step is $O(nd \log(n))$.

Computing $\hat{\Delta}$ requires computing the coherence $\hat{\Delta} = \max_{i \neq j} |e_i^\top S_f^\top S_f e_j|$. First, we note that if $S_f = \sqrt{\frac{n}{k}} P_{k \times n} H_n B_n$ we have $S_f^\top S_f = \frac{n}{k} B_n H_n^\top P_{k \times n}^\top P_{k \times n} H_n B_n$. Since B_n is diagonal with Rademacher entries, multiplying on the left and right by B_n only changes the signs of the entries, and therefore does not affect their absolute values. Hence,

$$\hat{\Delta} = \frac{n}{k} \max_{i \neq j} \left| e_i^\top H_n^\top P_{k \times n}^\top P_{k \times n} H_n e_j \right|.$$

Now let $z \in \{0, 1\}^n$ denote the indicator vector of the sampled rows, namely

$$z_r = 1 \quad \text{if row } r \text{ is selected by } P_{k \times n}, \quad z_r = 0 \text{ otherwise.}$$

Writing h_i for the i -th column of H_n , we obtain

$$e_i^\top H_n^\top P_{k \times n}^\top P_{k \times n} H_n e_j = \sum_{r=1}^n z_r (h_i)_r (h_j)_r.$$

Since $P_{k \times n} H_n$ is obtained by selecting k rows of the Hadamard matrix, and since the entry-wise product of two Hadamard columns is, up to the factor $1/\sqrt{n}$, another Hadamard column⁵, for every $i \neq j$ there exists a non-constant Hadamard column h_ℓ such that $h_i \odot h_j = \frac{1}{\sqrt{n}} h_\ell$ where $a \odot b$ denotes entry-wise product between the vectors a and b . Therefore,

$$e_i^\top H_n^\top P_{k \times n}^\top P_{k \times n} H_n e_j = \frac{1}{\sqrt{n}} \sum_{r=1}^n z_r (h_\ell)_r = \frac{1}{\sqrt{n}} (H_n z)_\ell.$$

It follows that

$$\hat{\Delta} = \frac{\sqrt{n}}{k} \max_{\ell \neq 0} |(H_n z)_\ell|,$$

where the maximum is taken over the values of the Hadamard transformed vector $H_n z$.

Hence, computing $\hat{\Delta}$ reduces to: (i) forming the indicator vector z of the sampled rows, (ii) applying one Hadamard transform to z , and (iii) taking the maximum absolute value over the resulting outputs. The overall complexity is therefore $O(n \log n)$ time and $O(n)$ memory.

⁵this holds since, by the definition of the elements of the Hadamard matrix it holds that $(H_n)_{i,j} (H_n)_{k,j} = (-1)^{\langle i,j \rangle_2 + \langle k,j \rangle_2} = (-1)^{\langle i \oplus k, j \rangle_2}$, corresponding to the element $(H_n)_{i \oplus k, j}$ and where $\langle a, b \rangle_2$ denotes the mod-2 inner product between the binary expansions of a and b

Computing $S_G Z$. Since the Gaussian sketch S_G is a dense matrix of size $k_1 \times k_2$, and since it is applied to a matrix of size $k_2 \times d$, this complexity is $O(k_1 k_2 d)$.

Overall Computation. Combining these terms leads to the next overall complexity

$$O(k_1 k_2 d + nd \log(n) + k_2 d^2 + d^3).$$

D Simplifying ϕ and Closed-Form (ε, δ) -DP Guarantees

Recall that we have established the upper bound

$$\max_{X' \simeq X} D_\alpha(\mathcal{M}_{S_f(X)} \| \mathcal{M}_{S_f(X')}) \leq \frac{1}{2(\alpha-1)} \cdot \left(\alpha \cdot \log \left(1 - \frac{1}{\gamma_f} - \frac{1}{4\gamma_f^2} \right) - \log \left(1 - \frac{\alpha}{\gamma_f} - \frac{\alpha^2}{4\gamma_f^2} \right) \right) := \bar{\phi}(\alpha; \gamma_f)$$

for all $1 < \alpha < 4\gamma_f/5$ and $\gamma_f > 5/4$ and where $\gamma_f := \frac{\lambda_{\min}^Z + \sigma^2}{C_{S_f X}^2}$.

Our goal now is to simplify $\bar{\phi}(\alpha; \gamma_f)$. As we will show, it admits a simplified expression which allows us to assess its degradation relative to the guarantees of the Gaussian sketch (given in Proposition 2). To that end, note that by Lev et al. [2025, Lemma. 1]

$$\max_{X' \simeq X} D_\alpha(\mathcal{M}_{\mathbb{I}_n(X)} \| \mathcal{M}_{\mathbb{I}_n(X')}) \leq \frac{1}{2(\alpha-1)} \cdot \left(\alpha \cdot \log \left(1 - \frac{1}{\gamma} \right) - \log \left(1 - \frac{\alpha}{\gamma} \right) \right) \quad (11)$$

for $\gamma > 1$ and $1 < \alpha < \gamma$ and where γ lower bounds $\lambda_{\min}^X + \sigma^2$. Additionally, by Lev et al. [2025, Corollary. 1], for all $1 < \alpha < 2\gamma/5$ it holds that

$$\frac{1}{2(\alpha-1)} \cdot \left(\alpha \cdot \log \left(1 - \frac{1}{\gamma} \right) - \log \left(1 - \frac{\alpha}{\gamma} \right) \right) \leq \frac{\alpha}{2\gamma^2}. \quad (12)$$

D.1 Upper Bound on $\bar{\phi}$

We note that in the range $1 < \alpha < 4\gamma_f/5$ and $\gamma_f > 5/4$ it holds that

$$\bar{\phi}(\alpha; \gamma_f) \leq \frac{1}{2(\alpha-1)} \cdot \left(\alpha \cdot \log \left(1 - \frac{5}{4\gamma_f} \right) - \log \left(1 - \frac{5\alpha}{4\gamma_f} \right) \right). \quad (13)$$

Then, using (12) we get that

$$\bar{\phi}(\alpha; \gamma_f) \leq \frac{25\alpha}{32\gamma_f^2}$$

for all $1 < \alpha < 8\gamma_f/25$. Since γ_f lower bounds $\frac{\lambda_{\min}^Z + \sigma^2}{C_{S_f X}^2}$, comparing (11) and (13) suggests that our new guarantees are degraded by the factor $\frac{5}{4} \cdot C_{S_f X}^2$ relative to Proposition 2, and moreover suffers from being dependent on the minimal eigenvalue $\lambda_{\min}^Z$ rather than on $\lambda_{\min}^X$. Since for many choices of S_f (for example, whenever S_f is an SRHT with $k\chi^2 = \Omega(\text{rank}(X) \cdot \log^4(n))$) it holds that $\lambda_{\min}^Z \geq (1 - \chi)\lambda_{\min}^X$ for $\chi \in (0, 1]$, this adds at most multiplicative degradation of $1 - \chi$.

D.2 Upper Bound on $\bar{\phi}$ Whenever $C_{S_f X}^2 \rightarrow \infty$

In the absence of any meaningful bound on $C_{S_f X}^2$, (namely, assuming that $C_{S_f X}^2 \rightarrow \infty$), by the analysis from Appendix C.2 the overall bound on the divergence becomes $\hat{\phi}(\alpha; \gamma_{f,\infty})$, where

$$\begin{aligned}\hat{\phi}(\alpha; \gamma_{f,\infty}) &= \frac{1}{2(\alpha-1)} \cdot \left(2\alpha \cdot \log \left(1 - \frac{1}{\sqrt{\gamma_{f,\infty}}} \right) - \log \left(1 - \frac{2\alpha}{\sqrt{\gamma_{f,\infty}}} + \frac{\alpha}{\gamma_{f,\infty}} \right) \right) \\ &= \frac{1}{2(\alpha-1)} \cdot \left(\alpha \cdot \log \left(1 - \left(\frac{2}{\sqrt{\gamma_{f,\infty}}} - \frac{1}{\gamma_{f,\infty}} \right) \right) - \log \left(1 - \alpha \left(\frac{2}{\sqrt{\gamma_{f,\infty}}} - \frac{1}{\gamma_{f,\infty}} \right) \right) \right)\end{aligned}$$

for all $1 < \alpha < \frac{\sqrt{\gamma_{f,\infty}}}{2}$ and provided that $\gamma_{f,\infty} > 1$ where $\gamma_{f,\infty}$ lower bounds $\lambda_{\min}^Z + \sigma^2$. Then, using (12) we get that this is upper bounded by

$$\hat{\phi}(\alpha; \gamma_{f,\infty}) \leq \frac{\alpha}{2} \cdot \left(\frac{2}{\sqrt{\gamma_{f,\infty}}} - \frac{1}{\gamma_{f,\infty}} \right)^2 = \frac{2\alpha}{\gamma_{f,\infty}} \cdot \left(1 - \frac{1}{2\sqrt{\gamma_{f,\infty}}} \right)^2 \leq \frac{2\alpha}{\gamma_{f,\infty}}$$

for all $1 < \alpha < \frac{\sqrt{\gamma_{f,\infty}}}{5}$, suggesting that this bound decays like $(\lambda_{\min}^Z + \sigma^2)^{-1}$ instead of $(\lambda_{\min}^Z + \sigma^2)^{-2}$ as in Lev et al. [2025]. Additionally, since it depends on $\lambda_{\min}^Z$, it is similarly degraded by the multiplicative constant $1 - \chi$.

D.3 Closed-Form (ε, δ) -DP Guarantees

Incorporating these inside the conversion from RDP to (ε, δ) -DP [Bun et al., 2018, Lemma. 6] and multiplying by k_1 , which amounts to applying a full Gaussian sketch (rather than a single Gaussian vector as currently analyzed), yields the closed-form upper bounds on the privacy guarantees, which are given by

$$\varepsilon \leq \begin{cases} \frac{2k_1}{\lambda_{\min}^Z + \sigma^2} + \sqrt{\frac{8k_1 \log(1/\delta)}{\lambda_{\min}^Z + \sigma^2}} & \text{if } \log(1/\delta) < \left(\frac{\sqrt{\lambda_{\min}^Z + \sigma^2}}{5} - 1 \right)^2 \cdot \frac{2k_1}{\lambda_{\min}^Z + \sigma^2} \\ \frac{2k_1}{5\sqrt{\lambda_{\min}^Z + \sigma^2}} + \frac{5 \log(1/\delta)}{\sqrt{\lambda_{\min}^Z + \sigma^2} - 5} & \text{otherwise} \end{cases}$$

if $C_{S_f}^2 \rightarrow \infty$ and by

$$\varepsilon \leq \begin{cases} \frac{25k_1}{32(\lambda_{\min}^Z + \sigma^2)^2} + \frac{\sqrt{50k_1 \log(1/\delta)}}{4 \cdot (\lambda_{\min}^Z + \sigma^2)} & \text{if } \log(1/\delta) < \left(\frac{8(\lambda_{\min}^Z + \sigma^2)}{25} - 1 \right)^2 \cdot \frac{25k_1}{32(\lambda_{\min}^Z + \sigma^2)^2} \\ \frac{k_1}{4 \cdot (\lambda_{\min}^Z + \sigma^2)} + \frac{\log(1/\delta)}{\frac{8}{25} \cdot (\lambda_{\min}^Z + \sigma^2) - 1} & \text{otherwise} \end{cases} \quad (14)$$

if $C_{S_f}^2 < \infty$.

E Proof of Theorem 2

Our proof follows similarly to the analysis of the iterative Hessian mixing, presented in Lev et al. [2026]. We first provide auxiliary lemmas that will assist us in establishing the accuracy guarantees.

E.1 Auxiliary Definitions and Lemmas

We first provide the RDP guarantees of the Gaussian mechanism, as established by [Mironov \[2017\]](#).

Lemma 7 ([\[Mironov, 2017\]](#)). *Let $f : \mathcal{X} \rightarrow \mathbb{R}^m$ be a map. If there exists $b > 0$ such for any $X, X' \in \mathcal{X}$ satisfying $X \simeq X'$ it holds that $\|f(X) - f(X')\| \leq b$, then the Gaussian mechanism $X \mapsto f(X) + \sigma\xi$ for $\xi \sim \mathcal{N}(\vec{0}_m, \mathbb{I}_m)$ satisfies $(\alpha, \frac{ab^2}{2\sigma^2})$ for all $\alpha > 1$.*

Next, we prove auxiliary lemmas for the overall accuracy guarantee. The first establishes guarantees for a sketch formed by concatenating Gaussian and fast sketches, as in [FastMix](#). We later use it to generalize [Lev et al. \[2026\]](#). The second uses [Cohen et al. \[2016, Theorem. 1\]](#) for bounding $\hat{\Delta}$. The third shows that the SRHT sketch preserves $\lambda_{\min}^X$ with high probability.

Lemma 8. *Let S_{tot} defined via $S_{\text{tot}} = S_G \hat{S}_H$ for $S_G \sim \mathcal{N}(0, \mathbb{I}_{k_1 \times (k_2+d)})$ and $\hat{S}_H := \begin{pmatrix} S_H & 0_{k_2 \times d} \\ 0_{d \times n} & \mathbb{I}_d \end{pmatrix}$ where $S_H \sim \text{SRHT}(k_2, n)$. Then, there exists universal constants (c_0, c_1, c_2, c_3) such that for any $A \in \mathbb{R}^{(n+d) \times d}$ of the form $A = \begin{bmatrix} B^\top & b\mathbb{I}_d \end{bmatrix}^\top$ for some $B \in \mathbb{R}^{n \times d}$ and $b \in \mathbb{R}$, and for any $\chi \in (0, 1]$ and $k_1, k_2 \in \mathbb{N}$ satisfying $k_1\chi^2 \geq c_0d$, $k_2\chi^2 \geq c_0 \cdot \text{rank}(B) \cdot \log^4(n)$ it holds*

$$\mathbb{P}(\mathcal{E}(A, \chi, k_1)) \geq 1 - c_1 \cdot \left(\exp\{-c_2 k_1 \chi^2\} + \exp\left\{-\frac{c_3 k_2 \chi^2}{\log^4(n)}\right\} \right)$$

where

$$\mathcal{E}(A, \chi, k_1) := \left\{ \sup_{u \in \mathbb{R}^d, v \in \mathbb{R}^d} \frac{|u^\top A^\top \left(\frac{1}{k_1} S_{\text{tot}}^\top S_{\text{tot}} - \mathbb{I}_{n+d} \right) Av|}{\|Au\| \cdot \|Av\|} \leq \frac{\chi}{2} \cap \inf_{v \in \mathbb{R}^d} \frac{\|S_{\text{tot}} Av\|^2}{k_1 \cdot \|Av\|^2} \geq 1 - \chi \right\}.$$

Proof. Let

$$E_H(\eta) := \left\{ \sup_{u \in \mathbb{R}^n, v \in \mathbb{R}^d} \frac{|u^\top (S_H^\top S_H - \mathbb{I}_n) Bv|}{\|u\| \cdot \|Bv\|} \leq \frac{\eta}{2}, \quad \inf_{v \in \mathbb{R}^d} \frac{\|S_H Bv\|^2}{\|Bv\|^2} \geq 1 - \eta \right\} \quad (15)$$

for $\eta \in (0, 1]$. By [Pilanci and Wainwright \[2016, Lemma. 1\]](#) applied as in [Pilanci and Wainwright \[2015, Corollary. 2\]](#), $\mathbb{P}(E_H(\eta)) \geq 1 - c_{1,H} \cdot \exp\{-c_{2,H} k_2 \eta^2\}$ provided that $k_2 \eta^2 \geq c_{0,H} \cdot \text{rank}(B) \cdot \log^4(n)$ for some universal constants $c_{0,H}, c_{1,H}, c_{2,H}$.⁶ In particular, this holds for the k_2 from the lemma statement and with $\eta \leftarrow \chi$ whenever $c_0 \geq c_{0,H}$.

Then, we note that conditioned on $E_H(\eta)$, for every $v \in \mathbb{R}^d$ it holds that

$$\|S_H Bv\|^2 + b^2 \|v\|^2 \geq (1 - \chi) \|Bv\|^2 + b^2 \|v\|^2 \geq (1 - \chi) (\|Bv\|^2 + b^2 \|v\|^2),$$

and which then implies

$$\inf_{v \in \mathbb{R}^d} \frac{\|S_H Bv\|^2 + b^2 \|v\|^2}{\|Bv\|^2 + b^2 \|v\|^2} \geq 1 - \chi. \quad (16)$$

⁶By our convention, sketches in the set $\text{SRHT}(k, n)$ contain additional normalization by factor $\sqrt{k}$ which was not presented in [Pilanci and Wainwright \[2016\]](#). Thus, we avoid the factor $1/k_2$ while considering the events in (15).

Additionally, we note that

$$\sup_{u_1 \in \mathbb{R}^n, u_2 \in \mathbb{R}^d, v \in \mathbb{R}^d} \frac{\left| (u_1^\top, u_2^\top) \left(\begin{pmatrix} \mathbf{S}_H^\top \mathbf{S}_H & 0_{n \times d} \\ 0_{d \times d} & \mathbb{I}_d \end{pmatrix} - \mathbb{I}_{n+d} \right) \begin{pmatrix} B \\ b\mathbb{I}_d \end{pmatrix} v \right|}{\sqrt{\|u_1\|^2 + \|u_2\|^2} \cdot \sqrt{\|Bv\|^2 + b^2 \|v\|^2}} \leq \sup_{u_1 \in \mathbb{R}^n, v \in \mathbb{R}^d} \frac{|u_1^\top (\mathbf{S}_H^\top \mathbf{S}_H - \mathbb{I}_n) Bv|}{\|u_1\| \cdot \|Bv\|} \leq \frac{\chi}{2}. \quad (17)$$

Now, we make the notation $\mathbf{C} := \widehat{\mathbf{S}}_H A \in \mathbb{R}^{k_2 \times d}$ and

$$E_G(\eta; M) := \left\{ \sup_{u \in \mathbb{R}^{k_2+d}, v \in \mathbb{R}^d} \frac{\left| u^\top \left(\frac{1}{k_1} \mathbf{S}_G^\top \mathbf{S}_G - \mathbb{I}_{k_2+d} \right) Mv \right|}{\|u\| \cdot \|Mv\|} \leq \frac{\eta}{2}, \quad \inf_{v \in \mathbb{R}^d} \frac{\|\mathbf{S}_G Mv\|^2}{k_1 \cdot \|Mv\|^2} \geq 1 - \eta \right\}. \quad (18)$$

Using [Pilanci and Wainwright \[2016, Lemma. 1\]](#), we know that for every realization $\mathbf{C} = C$ it holds that $\mathbb{P}(E_G(\eta; C)) \geq 1 - c_{1,G} \cdot \exp\{-c_{2,G} k_1 \eta^2\}$ provided that $k_1 \eta^2 \geq c_{0,G} \cdot \text{rank}(C)$, and which since $\text{rank}(C) \leq d$ holds by the constraints we have on k_1 in the lemma statement for $\eta \leftarrow \chi$. Then, by using the law of total probability, we note that under the same constraint on k_1 it further holds that

$$\mathbb{P}(E_G(\eta; \mathbf{C})) \geq 1 - c_{1,G} \cdot \exp\{-c_{2,G} k_1 \eta^2\}$$

Thus, since

$$\inf_{v \in \mathbb{R}^d} \frac{\|\mathbf{S}_G C v\|^2}{k_1 \cdot \|Av\|^2} = \inf_{v \in \mathbb{R}^d} \left\{ \frac{\|\mathbf{S}_G C v\|^2}{k_1 \cdot \|C v\|^2} \cdot \frac{\|C v\|^2}{\|Av\|^2} \right\},$$

we get that

$$\inf_{v \in \mathbb{R}^d} \frac{\|\mathbf{S}_G C v\|^2}{k_1 \cdot \|Av\|^2} \geq (1 - \chi) \cdot \inf_{v \in \mathbb{R}^d} \frac{\|C v\|^2}{\|Av\|^2} \geq (1 - \chi)^2$$

where the first inequality is by (18) and the second inequality is by (16), and both hold since they hold for every realization $\mathbf{C} = C$, thus holds with probability 1 whenever we look at the event with the random matrix $\mathbf{C}$.

Note that by picking $M \leftarrow \widehat{\mathbf{S}}_H A$ in (18) and for the k_1 in the lemma statement we have

$$\sup_{w \in \mathbb{R}^{k_2+d}, v \in \mathbb{R}^d} \frac{\left| w^\top \left(\frac{1}{k_1} \mathbf{S}_G^\top \mathbf{S}_G - \mathbb{I}_{k_2+d} \right) \widehat{\mathbf{S}}_H A v \right|}{\|w\| \cdot \|\widehat{\mathbf{S}}_H A v\|} \leq \frac{\chi}{2}.$$

Then, by writing

$$\frac{1}{k_1} \mathbf{S}_{\text{tot}}^\top \mathbf{S}_{\text{tot}} - \mathbb{I}_{n+d} = \frac{1}{k_1} \widehat{\mathbf{S}}_H^\top \mathbf{S}_G^\top \mathbf{S}_G \widehat{\mathbf{S}}_H - \mathbb{I}_{n+d} = \widehat{\mathbf{S}}_H^\top \left(\frac{1}{k_1} \mathbf{S}_G^\top \mathbf{S}_G - \mathbb{I}_{k_2+d} \right) \widehat{\mathbf{S}}_H + \left(\widehat{\mathbf{S}}_H^\top \widehat{\mathbf{S}}_H - \mathbb{I}_{n+d} \right) \quad (19)$$

and using the independence between $\mathbf{S}_H$ and $\mathbf{S}_G$ we note that for every $w, v \in \mathbb{R}^d$ and $u = Aw$ it holds that

$$\begin{aligned} \frac{\left| u^\top \left(\frac{1}{k_1} \mathbf{S}_{\text{tot}}^\top \mathbf{S}_{\text{tot}} - \mathbb{I}_{n+d} \right) Av \right|}{\|u\| \|Av\|} &= \frac{\left| \left(\widehat{\mathbf{S}}_H u \right)^\top \left(\frac{1}{k_1} \mathbf{S}_G^\top \mathbf{S}_G - \mathbb{I}_{k_2+d} \right) \widehat{\mathbf{S}}_H A v + u^\top \left(\widehat{\mathbf{S}}_H^\top \widehat{\mathbf{S}}_H - \mathbb{I}_{n+d} \right) Av \right|}{\|u\| \|Av\|} \\ &\leq \frac{\left| \left(\widehat{\mathbf{S}}_H u \right)^\top \left(\frac{1}{k_1} \mathbf{S}_G^\top \mathbf{S}_G - \mathbb{I}_{k_2+d} \right) \widehat{\mathbf{S}}_H A v \right| + \left| u^\top \left(\widehat{\mathbf{S}}_H^\top \widehat{\mathbf{S}}_H - \mathbb{I}_{n+d} \right) Av \right|}{\|u\| \|Av\|} \end{aligned}$$

$$\begin{aligned}
&\leq \frac{\frac{\chi}{2} \cdot \|\widehat{\mathbf{S}}_{\mathbf{H}} u\| \cdot \|\widehat{\mathbf{S}}_{\mathbf{H}} A v\| + \frac{\chi}{2} \cdot \|u\| \cdot \|A v\|}{\|u\| \|A v\|} \\
&= \frac{\frac{\chi}{2} \cdot \|\widehat{\mathbf{S}}_{\mathbf{H}} A w\| \cdot \|\widehat{\mathbf{S}}_{\mathbf{H}} A v\| + \frac{\chi}{2} \cdot \|A w\| \cdot \|A v\|}{\|A w\| \|A v\|} \\
&\leq \frac{1}{2} \left(\chi + \chi \cdot \left(1 + \frac{\chi}{2} \right) \right) \\
&\leq \frac{5}{4} \chi
\end{aligned}$$

where we have used (19) and (17) and the inequalities

$$\|\widehat{\mathbf{S}}_{\mathbf{H}} A v\|^2 \leq \left(1 + \frac{\chi}{2} \right) \|A v\|^2, \quad \|\widehat{\mathbf{S}}_{\mathbf{H}} A w\|^2 \leq \left(1 + \frac{\chi}{2} \right) \|A w\|^2$$

which due to (17). Since, as discussed, the conditional bound for $E_{\mathbf{G}}(\eta; \mathbf{C})$ holds uniformly for every $\mathbf{C} = C$ whenever $k_1 \eta_1^2 \geq c_{0,\mathbf{G}} d$, it holds that

$$\mathbb{P}(E_{\mathbf{G}}(\eta_1; C) \mid \mathbf{C} = C) \geq 1 - c_{1,\mathbf{G}} \cdot \exp \left\{ -c_{2,\mathbf{G}} k_1 \eta_1^2 \right\},$$

and therefore,

$$\begin{aligned}
\mathbb{P}(E_{\mathbf{H}}(\eta_2) \cap E_{\mathbf{G}}(\eta_1; \mathbf{C})) &= \mathbb{E} \left[\mathbb{1}_{E_{\mathbf{H}}(\eta_2)} \cdot \mathbb{P}(E_{\mathbf{G}}(\eta_1; \mathbf{C}) \mid \mathbf{C}) \right] \\
&\geq (1 - c_{1,\mathbf{G}} \cdot \exp \left\{ -c_{2,\mathbf{G}} k_1 \eta_1^2 \right\}) \cdot \mathbb{P}(E_{\mathbf{H}}(\eta_2)) \\
&\geq 1 - c_{1,\mathbf{G}} \cdot \exp \left\{ -c_{2,\mathbf{G}} k_1 \eta_1^2 \right\} - c_{1,\mathbf{F}} \cdot \exp \left\{ -\frac{c_{2,\mathbf{F}} k_2 \eta_2^2}{\log^4(n)} \right\}
\end{aligned}$$

where $\mathbb{1}_{E_{\mathbf{H}}(\eta)}$ denotes an indicator for the event $E_{\mathbf{H}}(\eta)$. The proof is finalized by adjusting constants and noting that this bound holds with $\eta_1 \leftarrow \chi, \eta_2 \leftarrow \chi$ for the k_1, k_2 used in the lemma statement. $\square$

The proof also implicitly justifies the choice of the parameters k_1 and k_2 in the lemma statement. Since the overall approximation budget χ is shared by the sketches $\mathbf{S}_{\mathbf{H}}$ and $\mathbf{S}_{\mathbf{G}}$, a natural choice is to split it evenly, setting the approximation parameter of each sketch to $\chi/2$, as in our calibration of k_1 and k_2 . Additionally, since k_2 relates to an SRHT sketch, it should be inflated by a factor $\log^4(n)$ to account for a similar factor in the constraints, and furthermore, since it is applied directly to B , it can be scaled with $\text{rank}(B)$ rather than d .

Lemma 9. *Let $\mathbf{S}_{\mathbf{H}} \sim \text{SRHT}(k, n)$ and let $X \in \mathbb{R}^{n \times d}$ with $n \geq d$. Then, there exist three universal constants $c_0, c_1, c_2 > 0$ such that for any $\chi, \varrho \in (0, 1]$ satisfying*

$$k \chi^2 \geq c_0 \cdot \left(\text{rank}(X) + \log \left(\frac{n c_1}{\chi \varrho} \right) \log \left(\frac{c_2 n (\text{rank}(X) + 1)}{\varrho} \right) \right),$$

then w.p. at least $1 - \varrho$,

$$\max_{i \in [n]} \left\| (\mathbf{S}_{\mathbf{H}} X)^\top \mathbf{S}_{\mathbf{H}}^\top e_i - x_i \right\| \leq \chi \|X\|_{\text{op}}.$$

Proof. Let $X = U \Sigma V^\top$ be a compact SVD of X , where $U \in \mathbb{R}^{n \times r}$ has orthonormal columns with $r := \text{rank}(X)$ and note that $\|X\|_{\text{op}} = \|V \Sigma\|_{\text{op}}$. Then, for every $i \in [n]$

$$\left\| (\mathbf{S}_{\mathbf{H}} X)^\top \mathbf{S}_{\mathbf{H}}^\top e_i - x_i \right\| = \left\| X^\top (\mathbf{S}_{\mathbf{H}}^\top \mathbf{S}_{\mathbf{H}} - \mathbb{I}_n) e_i \right\| = \left\| V \Sigma U^\top (\mathbf{S}_{\mathbf{H}}^\top \mathbf{S}_{\mathbf{H}} - \mathbb{I}_n) e_i \right\| \leq \|X\|_{\text{op}} \left\| U^\top (\mathbf{S}_{\mathbf{H}}^\top \mathbf{S}_{\mathbf{H}} - \mathbb{I}_n) e_i \right\|.$$

Thus, it is enough to bound $\|U^\top (\mathbf{S}_H^\top \mathbf{S}_H - \mathbb{I}_n) e_i\|$. Fix $i \in [n]$ and let $Q_i \in \mathbb{R}^{n \times t_i}$ have orthonormal columns spanning $\text{span}(U, e_i)$ and where $t_i \leq r + 1$. Since both U and e_i belong to $\text{span}(Q_i)$, there exist R_i and q_i such that

$$U = Q_i R_i, \quad e_i = Q_i q_i, \quad \|R_i\|_{\text{op}} \leq 1, \quad \|q_i\|_2 \leq 1.$$

Therefore,

$$\|U^\top (\mathbf{S}_H^\top \mathbf{S}_H - \mathbb{I}_n) e_i\| = \|R_i^\top Q_i^\top (\mathbf{S}_H^\top \mathbf{S}_H - \mathbb{I}_n) Q_i q_i\| \leq \|Q_i^\top (\mathbf{S}_H^\top \mathbf{S}_H - \mathbb{I}_n) Q_i\|_{\text{op}}.$$

Hence

$$\|X^\top (\mathbf{S}_H^\top \mathbf{S}_H - \mathbb{I}_n) e_i\| \leq \|X\|_{\text{op}} \|Q_i^\top (\mathbf{S}_H^\top \mathbf{S}_H - \mathbb{I}_n) Q_i\|_{\text{op}}.$$

Then, for each fixed i , by [Cohen et al. \[2016, Theorem. 9\]](#) it holds that

$$\mathbb{E} \left[\|Q_i^\top (\mathbf{S}_H^\top \mathbf{S}_H - \mathbb{I}_n) Q_i\|_{\text{op}}^2 \right] \leq \chi^2 \delta_i,$$

provided

$$k \geq \frac{C}{\chi^2} \left((r+1) + \log \left(\frac{1}{\chi \delta_i} \right) \log \left(\frac{r+1}{\delta_i} \right) \right)$$

for a universal constant $C > 0$. By Markov's inequality, this further implies that

$$\mathbb{P} \left(\|Q_i^\top (\mathbf{S}_H^\top \mathbf{S}_H - \mathbb{I}_n) Q_i\|_{\text{op}} > \chi \right) \leq \delta_i.$$

Consequently,

$$\mathbb{P} \left(\|X^\top (\mathbf{S}_H^\top \mathbf{S}_H - \mathbb{I}_n) e_i\| > \chi \|X\|_{\text{op}} \right) \leq \delta_i.$$

Then, choosing $\delta_i := \varrho/n$ and union bounding over $i \in [n]$ yields

$$\mathbb{P} \left(\max_{i \in [n]} \|X^\top (\mathbf{S}_H^\top \mathbf{S}_H - \mathbb{I}_n) e_i\| > \chi \|X\|_{\text{op}} \right) \leq \sum_{i=1}^n \delta_i = \varrho.$$

This completes the proof. $\square$

Lemma 10. Let $X \in \mathbb{R}^{n \times d}$ and let $\mathbf{S}_H \in \mathbb{R}^{k \times n}$ be SRHT sketch (Definition 3). There exists universal constants c_0, c_1, c_2 such that if $k \geq \frac{c_0 \cdot \text{rank}(X)}{\chi^2} \cdot \log^4(n)$ then w.p. at least $1 - c_1 \cdot \exp \left\{ -\frac{c_2 k \chi^2}{\log^4(n)} \right\}$

$$(1 - \chi) \cdot \lambda_{\min}^X \leq \lambda_{\min}^{\mathbf{S}_H X} \leq (1 + \chi) \cdot \lambda_{\min}^X, \quad (1 - \chi) \cdot \lambda_{\max}^X \leq \lambda_{\max}^{\mathbf{S}_H X} \leq (1 + \chi) \cdot \lambda_{\max}^X.$$

Proof. The proof follows by [Pilanci and Wainwright \[2016, Lemma. 1\]](#), which implies that for $k \geq \frac{c_0 \cdot \text{rank}(X)}{\chi^2} \cdot \log^4(n)$ w.p. at least $1 - c_1 \cdot \exp \left\{ -\frac{c_2 k \chi^2}{\log^4(n)} \right\}$

$$\sup_{v \in \mathbb{R}^d: \|v\|=1} \left| X^\top v^\top (\mathbf{S}_H^\top \mathbf{S}_H - \mathbb{I}_n) X v \right| \leq \chi \|X v\|^2. \quad (20)$$

We note that (20) implies

$$(1 - \chi) \|X v\|^2 \leq \|\mathbf{S}_H X v\|^2 \leq (1 + \chi) \|X v\|^2, \quad \forall v \in \mathbb{R}^d.$$

We now restrict attention to vectors v that have unit norm. Then, since by definition $\|Xv\|^2 \geq \lambda_{\min}^X$,

$$(1 - \chi) \lambda_{\min}^X \leq \|\mathbf{S}_H X v\|^2 \leq (1 + \chi) \|Xv\|^2.$$

Setting $v \leftarrow v^*$, where v^* corresponds to the minimal singular value of X , yields

$$(1 - \chi) \lambda_{\min}^X \leq \|\mathbf{S}_H X v^*\|^2 \leq (1 + \chi) \lambda_{\min}^X.$$

This implies

$$(1 - \chi) \lambda_{\min}^X \leq \lambda_{\min}^{\mathbf{S}_H X} \leq (1 + \chi) \lambda_{\min}^X$$

since $\|\mathbf{S}_H X v^*\|^2 \geq \lambda_{\min}^{\mathbf{S}_H X}$ and since we could instead pick v to be the singular vector corresponding to the minimum singular value of $\mathbf{S}_H X$. The corresponding derivation for the maximum eigenvalue proceeds in the same manner. The proof then follows after adjusting χ and the relevant constants. $\square$

Finally, the next lemma provides high probability bounds for the norm of a Gaussian vector

Lemma 11. *Let $\xi \sim \mathcal{N}(\vec{0}_d, \mathbb{I}_d)$ and let $\varrho \in (0, 1]$. Then,*

$$\mathbb{P}\left(\|\xi\|^2 \leq \left(\sqrt{2d} + \sqrt{4\log(1/\varrho)}\right)^2\right) \geq 1 - \varrho^2.$$

The previous lemmas and definitions allow us to derive high-probability accuracy guarantees for **FastIHM**. In particular, Lemma 8 parallels a similar lemma proved in Lev et al. [2026] and enables a recursive characterization. In addition, Lemma 9 and Lemma 10 control the degradation induced by applying the Gaussian sketch to the compressed datasets $\{\mathbf{S}_{H,t} X\}_t$, as done in **FastMix**. Using the notation of Lemma 1, they control $\lambda_{\min}^{\mathbf{S}_{H,t} X}$ and $C_{\mathbf{S}_{H,t} X}^2$ for all $t = 0, 1, \dots, T-1$.

E.2 Proof of Theorem 2

We first prove the DP guarantee for the algorithm, and then the utility guarantee. Throughout, given (k_1, T, C) and for a target failure probability ϱ we make the next choice of noise parameters

$$\sigma = \frac{3\sqrt{T \log(3/\delta)} C}{\sqrt{2}\varepsilon} \left(1 + \sqrt{1 + \frac{\varepsilon}{3 \log(3/\delta)}}\right), \quad \tau = \max \left\{ \log \left(\frac{3T}{\delta} \right), \log \left(\frac{16T}{\varrho} \right) \right\}, \quad \omega = \frac{6T}{\varepsilon}$$

$$\gamma_1 = \frac{15\sqrt{2k_1 T \log(3/\delta)}}{8\varepsilon} \left(1 + \sqrt{1 + \frac{\varepsilon}{3 \log(3/\delta)}}\right), \quad \gamma_2 = \frac{2k_1 T + 25 \log(3/\delta)}{8\varepsilon}$$

and

$$\gamma = \begin{cases} \gamma_1 & \text{if } \log(3/\delta) \leq \left(\frac{8\gamma_1}{25} - 1\right)^2 \cdot \frac{25k_1 T}{32\gamma_1^2} \\ \gamma_2 & \text{otherwise} \end{cases}.$$

E.2.1 Differential Privacy:

Algorithm 2 involves $2T$ calls to private mechanisms: T calls to the Gaussian mechanism for constructing $\tilde{G}_t$ and another T calls for forming $(\tilde{X}_t, \tilde{\eta}_t)$ (splitted between the function **MultipleFastMixing** and the fast sketching step in Line. 5). Inside the function **MultipleFastMixing**, $\tilde{m}_t$ and $\tilde{\lambda}_t$ are

computed via the Laplace mechanism. Moreover, given these private values, the RDP guarantees for the steps computing $\tilde{X}_t$ are known due to Lemma 1. Throughout, quantities computed in the t 'th call to **FastMix** are indexed by t (for example, $\tilde{m}_t, \tilde{\lambda}_t, \mathbf{S}_{H,t}, \mathbf{S}_{G,t}$).

Our proof uses Lemma 7 and Lemma 1 to calculate the RDP guarantees for each step forming $\tilde{G}_t$ and $\tilde{X}_t$, and then applies the RDP composition theorem to obtain their overall guarantees separately. In particular, we separate the contribution of the Laplace mechanisms from the overall RDP guarantees of the output of **FastMix**. We then convert these RDP guarantees to (ε, δ) -DP, and combine them with the ε -DP guarantee of the steps involving the Laplace mechanism.

Generating $\{\tilde{m}_t, \tilde{\lambda}_t\}_{t=0}^{T-1}$. As proved in Lemma 4, the ℓ_1 sensitivity of each $\tilde{m}_t$ is $\hat{\Delta}_t$, and as proved in Lemma 5, the ℓ_1 sensitivity of $\tilde{\lambda}_t$ is $\hat{\Lambda}_t + 2\tilde{m}_t$. Thus, under the event ensuring that $\tilde{m}_t$ provides an upper bound on $\hat{m}_t$, and since we use the Laplace mechanism for the release of these quantities, for the value of ω we use, the contribution of each of these steps is $\frac{\varepsilon}{6T}$, which after composing both steps over T iterations yields overall contribution of $\frac{\varepsilon}{3}$ to the overall ε .

Moreover, it holds that $\mathbb{P}(\tilde{m}_t \geq \hat{m}_t) = \mathbb{P}(\tilde{\lambda}_t \leq \lambda_{\min}^{\mathbf{S}_{H,t}X}) = 1 - \frac{1}{2} \cdot \exp\{-\tau\} \geq 1 - \frac{\delta}{6T}$. Thus, defining $\mathcal{B} := \bigcap_{t=0}^{T-1} \left\{ \{\tilde{m}_t \geq \hat{m}_t\} \cap \left\{ \tilde{\lambda}_t \leq \lambda_{\min}^{\mathbf{S}_{H,t}X} \right\} \right\}$, it holds that $\mathbb{P}(\mathcal{B}) \geq (1 - \frac{\delta}{6T})^{2T} \geq 1 - \frac{\delta}{3}$.

Gaussian Mechanism. Let $(X', Y') \simeq (X, Y)$ i.e., (X', Y') is a zero-out neighbor of (X, Y) . Then, since we set $C_X = 1$, by the Cauchy-Schwarz inequality we get that

$$\left\| X'^{\top} \text{clip}_C(Y' - X'\hat{\theta}_t) - X^{\top} \text{clip}_C(Y - X\hat{\theta}_t) \right\| \leq C \quad (21)$$

which holds since the clipping operation happens coordinate-wise, thus $X'^{\top} \text{clip}_C(Y' - X'\hat{\theta}_t) - X^{\top} \text{clip}_C(Y - X\hat{\theta}_t)$ is of the form $\text{clip}_C(y_i - x_i^{\top} \hat{\theta}_t) \cdot x_i$ for the coordinate i that have been removed and which by the assumption $\|x_i\| \leq 1$ is upper bounded by C . Thus, the ℓ_2 sensitivity of the quantity $X^{\top} \text{clip}_C(Y - X\hat{\theta}_t)$ is at most C irrespective of $\hat{\theta}_t$.

Then, Lemma 7 and (21) yield that the RDP guarantee of each step that forms $\tilde{G}_t$ is $\left(\alpha, \frac{\alpha C^2}{2\sigma^2}\right)$.

Fast Sketching. Under the event $\mathcal{B}$ it holds

$$\lambda_{\min}^{\mathbf{S}_{H,t}X} + \eta^2 \geq \gamma \cdot \left(1 + 2 \cdot \max_{i \in [n]} \left\| (\mathbf{S}_{H,t}X)^{\top} \mathbf{S}_{H,t} - x_i \right\| \right), \quad \text{for all } t = 0, 1, \dots, T-1$$

and for all the cases of the algorithm. Thus, by Lemma 1, the RDP guarantee of each of these steps is upper bounded by $(\alpha, k_1 \cdot \bar{\phi}(\alpha; \gamma))$ where $\bar{\phi}(\alpha; \gamma)$ was defined in Section 3 and is given by

$$\bar{\phi}(\alpha; \gamma) := \frac{1}{2(\alpha - 1)} \cdot \left(\alpha \cdot \log \left(1 - \frac{1}{\gamma} - \frac{1}{4\gamma^2} \right) - \log \left(1 - \frac{\alpha}{\gamma} - \frac{\alpha^2}{4\gamma^2} \right) \right) \quad \text{for } \gamma > 5/4 \text{ and } \alpha \in (1, 4\gamma/5).$$

Composition. We compose the T different steps of the Gaussian mechanism and the T sketching steps in RDP. Then, we convert this to (ε, δ) -DP via the conversion from Canonne et al. [2020], and then compose with the steps that perform the calculations of $\tilde{m}_t$ and $\tilde{\lambda}_t$ in (ε, δ) -DP. For the T steps of the Gaussian mechanism, we get

$$\hat{\varepsilon}_{\text{Gauss}} = \min_{\alpha > 1} \left\{ \frac{\alpha T C^2}{2\sigma^2} + \frac{\log(3/\delta) + (\alpha - 1) \log(1 - 1/\alpha) - \log(\alpha)}{\alpha - 1} \right\}$$

$$\begin{aligned}
&\leq \min_{\alpha > 1} \left\{ \frac{\alpha TC^2}{2\sigma^2} + \frac{\log(3/\delta)}{\alpha - 1} \right\} \\
&= \frac{\sqrt{2T \log(3/\delta)} C}{\sigma} + \frac{TC^2}{2\sigma^2}
\end{aligned}$$

and $\widehat{\delta}_{\text{Gauss}} = \delta/3$. For the T sketching steps, we get

$$\widehat{\varepsilon}_{\text{FastMix}} = \min_{1 < \alpha < 4\gamma/5} \left\{ k_1 T \cdot \bar{\phi}(\alpha; \gamma) + \frac{\log(3/\delta) + (\alpha - 1) \log(1 - 1/\alpha) - \log(\alpha)}{\alpha - 1} \right\}$$

and $\widehat{\delta}_{\text{FastMix}} = \delta/3$. Then, we get that the overall process is $(\widehat{\varepsilon}, \widehat{\delta})$ -DP where

$$\widehat{\delta} = \frac{\delta}{3} + \widehat{\delta}_{\text{FastMix}} + \widehat{\delta}_{\text{Gauss}} = \delta$$

where the first $\delta/3$ is by the probability of the event $\mathcal{B}$ not holding (which, according to Lemma 6 added as an additional term to δ) and where $\widehat{\varepsilon}$ is

$$\begin{aligned}
\widehat{\varepsilon} &\leq \frac{\varepsilon}{3} + \widehat{\varepsilon}_{\text{FastMix}} + \widehat{\varepsilon}_{\text{Gauss}} \\
&\leq \frac{\varepsilon}{3} + \min_{1 < \alpha < 4\gamma/5} \left\{ k_1 T \cdot \bar{\phi}(\alpha; \gamma) + \frac{\log(3/\delta) + (\alpha - 1) \log(1 - 1/\alpha) - \log(\alpha)}{\alpha - 1} \right\} \\
&\quad + \frac{\sqrt{2T \log(3/\delta)} C}{\sigma} + \frac{TC^2}{2\sigma^2} \\
&\leq \frac{\varepsilon}{3} + \min_{1 < \alpha < 4\gamma/5} \left\{ k_1 T \cdot \bar{\phi}(\alpha; \gamma) + \frac{\log(3/\delta)}{\alpha - 1} \right\} \\
&\quad + \frac{\sqrt{2T \log(3/\delta)} C}{\sigma} + \frac{TC^2}{2\sigma^2}.
\end{aligned}$$

Now, for our choice of σ , the sum of the third and fourth terms equals $\varepsilon/3$. Moreover, by (14) in Appendix D, for our choice of γ the second term is upper bounded by $\varepsilon/3$. Combining these bounds with Lemma 6 yields $\widehat{\varepsilon} = \varepsilon$ and concludes the proof. $\square$

E.2.2 Excess Empirical Risk:

We follow the proof suggested in Lev et al. [2026] and seek to obtain an high probability bound on $\|\widehat{\theta}_T - \theta^*\|_{X^\top X}^2$, which corresponds to the excess empirical risk [Wang, 2018, Lemma. 5].

Recall that in terms of notation, $\widetilde{m}_t$ denote the private version of $\widehat{m}_t$ and $\widetilde{\lambda}_t$ the private version of $\widehat{\lambda}_t$, where $\widehat{\lambda}_t := \lambda_{\min}^{\text{S}_{H,t} X}$. For any $a \geq 0$, we define $\overline{X}_a := (X^\top, a\mathbb{I}_d)^\top$ and $M_a := X^\top X + a^2 \mathbb{I}_d = \overline{X}_a^\top \overline{X}_a$.

Step 1: Defining the High-Probability Event.

We perform the analysis on an explicit high-probability event. For each $t = 0, \dots, T-1$, define

$$\begin{aligned}
\mathcal{B}_{1,t}(\chi) &:= \left\{ \max_{i \neq j} \left| e_i^\top \text{S}_{H,t}^\top \text{S}_{H,t} e_j \right| \leq \chi \right\}, \\
\mathcal{B}_{2,t}(\chi) &:= \left\{ \sup_{u \in \mathbb{R}^n, v \in \mathbb{R}^d} \frac{\left| u^\top \left(\text{S}_{H,t}^\top \text{S}_{H,t} - \mathbb{I}_n \right) X v \right|}{\|u\| \|X v\|} \leq \frac{\chi}{2}, \inf_{v \in \mathbb{R}^d} \frac{\|\text{S}_{H,t} X v\|^2}{\|X v\|^2} \geq 1 - \chi \right\}.
\end{aligned}$$

By [Pilanci and Wainwright \[2016, Lemma. 1\]](#), there exist constants $C_H, c_H, \bar{c}_H$ such that for any $\chi \in (0, 1]$, $\mathbb{P}(\mathcal{B}_{2,t}(\chi)) \geq 1 - \bar{c}_H \cdot \exp\left\{-\frac{c_H k_2 \chi^2}{\log^4(n)}\right\}$ provided that $k_2 \chi^2 \geq C_H \cdot \text{rank}(X) \cdot \log^4(n)$.

Moreover, using $X = e_j$ for all $j \in [n]$ in Lemma 9, there exist $C_{\text{coh}}, c_{\text{coh}}, \bar{c}_{\text{coh}} > 0$ such that $\mathbb{P}(\mathcal{B}_{1,t}(\chi)) \geq 1 - \frac{\varrho}{4T}$ provided that $k_2 \chi^2 \geq C_{\text{coh}} \left(1 + \log\left(\frac{\bar{c}_{\text{coh}} T n^2}{\chi \varrho}\right) \log\left(\frac{c_{\text{coh}} T n^2}{\varrho}\right)\right)$. Therefore, if

$$k_2 \chi^2 \geq \max \left\{ C_H \cdot \text{rank}(X) \cdot \log^4(n), \frac{\log^4(n)}{c_H} \log\left(\frac{4T \bar{c}_H}{\varrho}\right), C_{\text{coh}} \left(1 + \log\left(\frac{\bar{c}_{\text{coh}} T n^2}{\chi \varrho}\right) \log\left(\frac{c_{\text{coh}} T n^2}{\varrho}\right)\right) \right\},$$

then $\mathbb{P}(\mathcal{B}_{1,t}^c(\chi)) \leq \frac{\varrho}{4T}$ and $\mathbb{P}(\mathcal{B}_{2,t}^c(\chi)) \leq \frac{\varrho}{4T}$ for every $t = 0, 1, \dots, T-1$. Now, define

$$\mathcal{B}_{\text{tot}}(\chi) := \bigcap_{t=0}^{T-1} (\mathcal{B}_{1,t}(\chi) \cap \mathcal{B}_{2,t}(\chi)).$$

By the union bound,

$$\mathbb{P}(\mathcal{B}_{\text{tot}}^c(\chi)) \leq \sum_{t=0}^{T-1} \mathbb{P}(\mathcal{B}_{1,t}^c(\chi)) + \sum_{t=0}^{T-1} \mathbb{P}(\mathcal{B}_{2,t}^c(\chi)) \leq T \cdot \frac{\varrho}{4T} + T \cdot \frac{\varrho}{4T} = \frac{\varrho}{2},$$

and hence $\mathbb{P}(\mathcal{B}_{\text{tot}}(\chi)) \geq 1 - \frac{\varrho}{2}$. We note that the event $\mathcal{B}_{\text{tot}}(\chi)$ implies

$$\lambda_{\min}^{\mathbf{S}_{H,t} X} \geq (1 - \chi) \lambda_{\min}^X \quad \text{and} \quad \max_{i \in [n]} \left\| X^\top \left(\mathbf{S}_{H,t}^\top \mathbf{S}_{H,t} - \mathbb{I}_n \right) e_i \right\| \leq \frac{\chi}{2} \|X\|_{\text{op}}$$

for all $t = 0, \dots, T-1$. This holds since for every $i \in [n]$, the event $\mathcal{B}_{2,t}(\chi)$ implies

$$\left\| X^\top \left(\mathbf{S}_{H,t}^\top \mathbf{S}_{H,t} - \mathbb{I}_n \right) e_i \right\| = \sup_{\|v\|=1} \left| e_i^\top \left(\mathbf{S}_{H,t}^\top \mathbf{S}_{H,t} - \mathbb{I}_n \right) X v \right| \leq \frac{\chi}{2} \sup_{\|v\|=1} \|X v\| = \frac{\chi}{2} \|X\|_{\text{op}}.$$

Conditioning on $\mathcal{B}_{\text{tot}}(\chi)$, we next define for each $t = 0, \dots, T-1$ the events

$$\mathcal{A}_{1,t}(\chi) := \left\{ \sup_{u \in \mathbb{R}^{k_2+d}, v \in \mathbb{R}^d} \frac{\left| u^\top \left(\frac{1}{k_1} (\mathbf{S}_{G,t}, \xi_t)^\top (\mathbf{S}_{G,t}, \xi_t) - \mathbb{I}_{k_2+d} \right) \begin{pmatrix} \mathbf{S}_{H,t} X \\ \eta \mathbb{I}_d \end{pmatrix} v \right|}{\|u\| \left\| \begin{pmatrix} \mathbf{S}_{H,t} X \\ \eta \mathbb{I}_d \end{pmatrix} v \right\|} \leq \frac{\chi}{2}, \inf_{v \in \mathbb{R}^d} \frac{\left\| (\mathbf{S}_{G,t}, \xi_t) \begin{pmatrix} \mathbf{S}_{H,t} X \\ \eta \mathbb{I}_d \end{pmatrix} v \right\|^2}{k_1 \left\| \begin{pmatrix} \mathbf{S}_{H,t} X \\ \eta \mathbb{I}_d \end{pmatrix} v \right\|^2} \geq 1 - \chi \right\},$$

$$\mathcal{A}_{2,t} := \left\{ \hat{m}_t \leq \tilde{m}_t \leq \hat{m}_t + \frac{12\chi T \max\{\log(3T/\delta), \log(16T/\varrho)\}}{\varepsilon} \right\}, \quad \mathcal{A}_{3,t} := \{\mathbf{z}_{2,t} \leq \tau\},$$

$$\mathcal{A}_{4,t} := \left\{ \|\zeta_t\|^2 \leq \left(\sqrt{2d} + \sqrt{4 \log(8T/\varrho)} \right)^2 \right\}.$$

By [Pilanci and Wainwright \[2016, Lemma. 1\]](#), there exist constants $C_G, c_G, \bar{c}_G > 0$ such that $\mathcal{A}_{1,t}(\chi)$ holds w.p. at least $1 - \frac{\varrho}{8T}$ provided that $k_1 \chi^2 \geq \max \left\{ C_G d, \frac{1}{c_G} \log\left(\frac{8T \bar{c}_G}{\varrho}\right) \right\}$. Moreover, by Lemma 11 applied with $\xi \leftarrow \zeta_t$ and $\varrho \leftarrow \varrho/8T$ it holds that

$$\mathbb{P}(\mathcal{A}_{4,t}^c) \leq \left(\frac{\varrho}{8T} \right)^2 \leq \frac{\varrho}{8T},$$

and by our choice of τ and since $\mathbf{z}_{2,t} \sim \text{Lap}(0, 1)$,

$$\mathbb{P}(\mathcal{A}_{3,t}^c) \leq \frac{\varrho}{8T}.$$

Finally, under $\mathcal{B}_{\text{tot}}$ and for our choice of τ ,

$$\mathbb{P}(\mathcal{A}_{2,t}^c \mid \mathcal{B}_{\text{tot}}(\chi)) \leq \frac{\varrho}{8T}$$

which holds since conditioned on $\mathcal{B}_{\text{tot}}(\chi)$, $\mathcal{A}_{2,t}$ is implied by the event $\{\mathbf{z}_{1,t} \in [-\tau, \tau]\}$. Now, define

$$\mathcal{A}_{\text{tot}}(\chi) := \mathcal{B}_{\text{tot}}(\chi) \cap \bigcap_{t=0}^{T-1} (\mathcal{A}_{1,t}(\chi) \cap \mathcal{A}_{2,t} \cap \mathcal{A}_{3,t} \cap \mathcal{A}_{4,t}).$$

Then,

$$\begin{aligned} \mathbb{P}(\mathcal{A}_{\text{tot}}^c(\chi)) &\leq \mathbb{P}(\mathcal{B}_{\text{tot}}^c(\chi)) + \mathbb{P}\left(\bigcup_{t=0}^{T-1} (\mathcal{A}_{1,t}^c(\chi) \cup \mathcal{A}_{2,t}^c \cup \mathcal{A}_{3,t}^c \cup \mathcal{A}_{4,t}^c) \mid \mathcal{B}_{\text{tot}}(\chi)\right) \\ &\leq \frac{\varrho}{2} + \sum_{t=0}^{T-1} \left(\frac{\varrho}{8T} + \frac{\varrho}{8T} + \frac{\varrho}{8T} + \frac{\varrho}{8T} \right) \\ &= \varrho \end{aligned}$$

and which implies $\mathbb{P}(\mathcal{A}_{\text{tot}}(\chi)) \geq 1 - \varrho$, provided that

$$\begin{aligned} k_1 \chi^2 &\geq \max \left\{ C_G d, \frac{1}{c_G} \log \left(\frac{8T \bar{c}_G}{\varrho} \right) \right\}, \\ k_2 \chi^2 &\geq \max \left\{ C_H \cdot \text{rank}(X) \cdot \log^4(n), \frac{\log^4(n)}{c_H} \log \left(\frac{4T \bar{c}_H}{\varrho} \right), C_{\text{coh}} \left(2 + \log \left(\frac{\bar{c}_{\text{coh}} T n^2}{\chi \varrho} \right) \log \left(\frac{c_{\text{coh}} T n^2}{\varrho} \right) \right) \right\}, \end{aligned}$$

for constants $(C_G, c_G, \bar{c}_G, C_H, c_H, \bar{c}_H, C_{\text{coh}}, c_{\text{coh}}, \bar{c}_{\text{coh}})$ and $\chi, \varrho \in (0, 1]$.

Step 2: Extending Auxiliary Lemmas From [Lev et al. \[2026\]](#):

Recall that Lemma 8 extends [Lev et al. \[2026, Lemma. 8\]](#) by showing that the concatenated sketch satisfies $\mathcal{C}(A, \chi, k)$ for a matrix A , approximation parameter χ , and sketch dimension k . It remains to extend [Lev et al. \[2026, Lemmas 10–12\]](#). For completeness, we restate these lemmas in terms of $\mathcal{C}(A, \chi, k)$ and derive them from [Lev et al. \[2026\]](#). We begin by recalling the definition of $\mathcal{C}(A, \chi, k)$:

$$\mathcal{C}(A, \chi, k) := \left\{ S \in \mathbb{R}^{k \times n} : \sup_{u \in \mathbb{R}^d, v \in \mathbb{R}^d} \frac{|u^\top A^\top (\frac{1}{k} S^\top S - \mathbb{I}_n) A v|}{\|A u\| \cdot \|A v\|} \leq \frac{\chi}{2} \bigcap \inf_{v \in \mathbb{R}^d} \frac{\|S A v\|^2}{k \cdot \|A v\|^2} \geq 1 - \chi \right\}.$$

Lemma 12. (*Extension of [Lev et al. \[2026, Lemma. 10\]](#). Given $\chi \in (0, 1/2]$, $A \in \mathbb{R}^{n \times d}$, $k \in \mathbb{N}$, $B \in \mathcal{C}(A, \chi, k)$, $b \in \mathbb{R}^n$ and $\theta_0 \in \mathbb{R}^d$ we have*

$$\left\| \widehat{\theta}(A, b, B, \theta_0, \vec{0}_d) - \theta^*(A, b) \right\|_{A^\top A}^2 \leq \chi^2 \|\theta_0 - \theta^*(A, b)\|_{A^\top A}^2.$$

Proof. As in [Lev et al. \[2026\]](#), the result follows directly from the analysis in [Pilanci and Wainwright \[2016, Appendix. C\]](#), with θ_0 playing the role of x^t in [Pilanci and Wainwright \[2016, Equation. 42\]](#) and B playing the role of the sketch S . The desired guarantee then follows from the assumption $B \in \mathcal{C}(A, \chi, k)$, since the event in [Pilanci and Wainwright \[2016, Equation. 43\]](#) corresponds to the setting of $\mathcal{C}$, where the left and right vectors multiplying $\frac{1}{k}B^\top B - \mathbb{I}_n$ are of the form Av . Hence, [Pilanci and Wainwright \[2016, Equation. 44a\]](#) holds under events of the form $B \in \mathcal{C}(A, \chi, k)$. $\square$

Lemma 13. (*Extension of [Lev et al. \[2026, Lemma. 11\]](#)*). Given $\chi \in (0, 1]$, $A \in \mathbb{R}^{n \times d}$ and $B \in \mathbb{R}^{k \times n}$ such that $B \in \mathcal{C}(A, \chi, k)$ and $A^\top A \succ 0$ it holds

$$\frac{1}{1 + \frac{\chi}{2}} (A^\top A)^{-1} \preceq \left(\frac{1}{k} (BA)^\top (BA) \right)^{-1} \preceq \frac{1}{1 - \frac{\chi}{2}} (A^\top A)^{-1}$$

and furthermore for any vector $v \in \mathbb{R}^d$

$$k^2 \left\| \left((BA)^\top (BA) \right)^{-1} v \right\|_{A^\top A}^2 \leq \frac{1}{(1 - \frac{\chi}{2})^2} \cdot \frac{\|v\|^2}{\lambda_{\min}(A^\top A)}.$$

Proof. The proof follows [Lev et al. \[2026, Lemma. 11\]](#), noting that [Lev et al. \[2026, Equation. 23\]](#) remains valid under the event $B \in \mathcal{C}(A, \chi, k)$, since it only requires multiplying $\frac{1}{k}B^\top B$ from both sides by a vector of the form Av , thus covered by $\mathcal{C}(A, \chi, k)$. The rest of the proof is unchanged. $\square$

Lemma 14. (*Extension of [Lev et al. \[2026, Lemma. 12\]](#)*). Let $Y \in \mathbb{R}^n$ and $X \in \mathbb{R}^{n \times d}$ satisfying $C_X = 1$, and let $M := X^\top X$. Suppose $\{B_t\}_{t=0}^{T-1}$ and $\{v_t\}_{t=0}^{T-1}$ satisfy $B_t \in \mathcal{C}(X, \chi, k)$ and $v_t \in \mathcal{A}_4(\varrho, d)$ for all $t \in \{0, \dots, T-1\}$ and for parameters $\chi, \varrho \in (0, 1]$ and $k \in \mathbb{N}$. Assume that $\frac{\chi \sqrt{\kappa(M)}}{2 - \chi} < 1$ where $\kappa(M) := \frac{\lambda_{\max}(M)}{\lambda_{\min}(M)}$. If $\{\hat{\theta}_t\}_{t=0}^{T-1}$ is defined according to the recursion

$$\hat{\theta}_{t+1} \left(\hat{\theta}_t, \sigma v_t \right) = \hat{\theta}_t + \left(\frac{1}{k} X^\top B_t^\top B_t X \right)^{-1} \left(X^\top (Y - X \hat{\theta}_t) + \sigma v_t \right), \quad \hat{\theta}_0 = \vec{0}_d,$$

for $\sigma \geq 0$, then

$$\|Y - X \hat{\theta}_t\|_\infty \leq C_Y + \left(1 + \left(\frac{\chi \sqrt{\kappa(M)}}{2 - \chi} \right)^t \right) \|\theta^*\| + \frac{2\sigma}{2 - \chi(1 + \sqrt{\kappa(M)})} \cdot \frac{\sqrt{2d} + \sqrt{4 \log(8T/\varrho)}}{\lambda_{\min}(M)}.$$

Proof. The proof follows since [Lemma 12](#) and [Lemma 13](#) implies that [Lev et al. \[2026, Lemma. 11\]](#) and [Lev et al. \[2026, Lemma. 10\]](#) holds. The remaining steps follow as in [Lev et al. \[2026\]](#). $\square$

Step 3: Adjusting the Overall Guarantee From [Lev et al. \[2026\]](#):

Throughout, we denote $\bar{X}_\eta := (X^\top, \eta \mathbb{I}_d)^\top$. Before presenting the proof we recall that for any $\chi \in (0, 1]$, conditioned on $\mathcal{A}_{\text{tot}}(\chi)$ it holds that

$$(1 - \chi) \lambda_{\min}^X \leq \hat{\lambda}_t \leq (1 + \chi) \lambda_{\min}^X, \quad \hat{\Delta}_t \leq \chi, \quad \hat{m}_t \leq \chi \sqrt{\lambda_{\max}^X} \quad (22)$$

and $\tilde{\lambda}_t \leq \hat{\lambda}_t$, $\hat{m}_t \leq \tilde{m}_t \leq \hat{m}_t + \frac{12\chi T \cdot \max\{\log(3T/\delta), \log(16T/\varrho)\}}{\varepsilon}$, all holds $\forall t = 0, 1, \dots, T-1$. Consequently,

$$\bar{m} := \max_{t=0,1,\dots,T-1} \tilde{m}_t \leq \chi \left(\sqrt{\lambda_{\max}^X} + \frac{12T \cdot \max\{\log(3T/\delta), \log(16T/\varrho)\}}{\varepsilon} \right).$$

Additionally, $\mathcal{A}_{\text{tot}}(\chi)$ holds under the next choice of parameters

$$\begin{aligned} k_1 \chi^2 &\geq \max \left\{ C_G d, \frac{1}{c_G} \log \left(\frac{8T \bar{c}_G}{\varrho} \right) \right\}, \\ k_2 \chi^2 &\geq \max \left\{ C_H \cdot \text{rank}(X) \cdot \log^4(n), \frac{\log^4(n)}{c_H} \log \left(\frac{4T \bar{c}_H}{\varrho} \right), C_{\text{coh}} \left(2 + \log \left(\frac{\bar{c}_{\text{coh}} T n^2}{\chi \varrho} \right) \log \left(\frac{c_{\text{coh}} T n^2}{\varrho} \right) \right) \right\}. \end{aligned} \quad (23)$$

Lastly, we assume that $\gamma = \gamma_1$, which puts the additional constraint $k_1 T = \Omega(\log(1/\delta))$.

Starting with the no-clipping conditions (paralleling [Lev et al. \[2026, Appendix . C, Step. 2\]](#)), we follow the notation from [Lev et al. \[2026, Appendix. C\]](#) and define $\kappa^X(a) := \frac{\lambda_{\max}^X + a}{\lambda_{\min}^X + a}$, $\kappa(A) := \frac{\lambda_{\max}(A)}{\lambda_{\min}(A)}$. First, for $t = 0$ no clipping occurs whenever $C \geq C_Y$. Then, for every $t \geq 1$, given $\chi \in (0, 1]$ satisfying $\frac{\chi \sqrt{\kappa^X(\eta^2)}}{2 - \chi} < 1$, by [Lemma 14](#) and [Lev et al. \[2026, Appendix. C, Step 2\]](#)

$$\begin{aligned} \|Y - X \hat{\theta}_t\|_{\infty} &\leq C_Y + \|\theta^*\| + \frac{1}{1 - \frac{\chi \sqrt{\kappa^X(\eta^2)}}{2 - \chi}} \cdot \frac{\sigma}{1 - \frac{\chi}{2}} \cdot \frac{\sqrt{2d} + \sqrt{4 \log(8T/\varrho)}}{\lambda_{\min}^X + \eta^2} + \left(\frac{\chi \sqrt{\kappa^X(\eta^2)}}{2 - \chi} \right)^t \|\theta^*\| \\ &\stackrel{(a)}{\leq} C_Y + \|\theta^*\| + \frac{1}{1 - \frac{\chi \sqrt{\kappa^X(\eta^2)}}{2 - \chi}} \cdot \frac{\sigma}{1 - \frac{\chi}{2}} \cdot \frac{\sqrt{2d} + \sqrt{4 \log(8T/\varrho)}}{(1 - \chi) \gamma} + \frac{\chi \sqrt{\kappa^X(\eta^2)}}{2 - \chi} \|\theta^*\| \\ &\stackrel{(b)}{\leq} C_Y + \|\theta^*\| + \frac{1}{1 - \frac{\chi \sqrt{\kappa^X(\eta^2)}}{2 - \chi}} \cdot \frac{C}{(1 - \chi)^2} \cdot \frac{\sqrt{2d} + \sqrt{4 \log(8T/\varrho)}}{\sqrt{k_1}} + \frac{\chi \sqrt{\kappa^X(\eta^2)}}{2 - \chi} \|\theta^*\| \\ &\stackrel{(c)}{\leq} C_Y + \|\theta^*\| + \frac{\chi}{1 - \frac{\chi \sqrt{\kappa^X(\eta^2)}}{2 - \chi}} \cdot \frac{C}{(1 - \chi)^2} + \frac{\chi \sqrt{\kappa^X(\eta^2)}}{2 - \chi} \|\theta^*\| \end{aligned}$$

where (a) is due to [\(22\)](#), since

$$(1 - \chi) \gamma \leq (1 - \chi) (\hat{\lambda}_t + \eta^2) \leq (1 - \chi) \hat{\lambda}_t + \eta^2 \leq \frac{\hat{\lambda}_t}{1 + \chi} + \eta^2 \leq \lambda_{\min}^X + \eta^2, \quad (24)$$

(b) is by substituting γ and σ , since $\frac{1 + \sqrt{1 + \frac{\sigma}{4}}}{1 + \sqrt{1 + \frac{\sigma}{4}}} \leq 2$, and (c) holds by [\(23\)](#), for sufficiently large (yet still constants) C_G , $\frac{1}{c_G}$ and $\bar{c}_G$. This is below C whenever

$$\frac{C_Y}{1 - \frac{\chi}{\left(1 - \frac{\chi \sqrt{\kappa^X(\eta^2)}}{2 - \chi}\right)(1 - \chi)^2}} + \frac{1 + \frac{\chi \sqrt{\kappa^X(\eta^2)}}{2 - \chi}}{1 - \frac{\chi}{\left(1 - \frac{\chi \sqrt{\kappa^X(\eta^2)}}{2 - \chi}\right)(1 - \chi)^2}} \|\theta^*\| \leq C$$

provided that $\frac{\chi}{\left(1 - \frac{\chi \sqrt{\kappa^X(\eta^2)}}{2 - \chi}\right)(1 - \chi)^2} < 1$. As in [Lev et al. \[2026, Appendix. C\]](#), for χ below a sufficiently

small constant less than $\frac{1}{4}$, attainable by increasing k_1 and k_2 by constant factors, this bound is less than C whenever $C \geq a_0 \cdot \max\{C_Y, \|\theta^*\|\}$ and $k_1 \geq a_1 \cdot \kappa^X(\eta^2) \cdot d$, $k_2 \geq a_1 \cdot \kappa^X(\eta^2) \cdot d \cdot \log^4(n)$ for constants a_0, a_1 . Now, to further simplify this argument, we note that

$$\kappa^X(\eta^2) = \frac{\lambda_{\max}^X - \lambda_{\min}^X}{\lambda_{\min}^X + \eta^2} + 1 \leq \frac{\lambda_{\max}^X - \lambda_{\min}^X}{(1 - \chi) \gamma} + 1$$

which holds by using (24). Then, provided that $\chi \in (0, 1/4]$, following analysis similar to that presented in Lev et al. [2026] we get that the constraints on k_1, k_2 can be enforced by setting

$$k_1 \geq a_2 \cdot \left(d \cdot \frac{(\lambda_{\max}^X - \lambda_{\min}^X) \varepsilon}{\sqrt{\log(1/\delta)}} \right)^{2/3}, \quad k_2 \geq a_2 \cdot \left(d \cdot \frac{(\lambda_{\max}^X - \lambda_{\min}^X) \varepsilon}{\sqrt{\log(1/\delta)}} \cdot \log^4(n) \right)^{2/3}.$$

Overall, conditioned on $\mathcal{A}_{\text{tot}}$ clipping does not occur provided that $C \geq a_0 \cdot \max\{C_Y, \|\theta^*\|\}$ and

$$k_1 \chi^2 \geq a_2 \cdot \max \left\{ d, g^X, \log \left(\frac{8T\bar{c}_G}{\varrho} \right), \left(\sqrt{2d} + \sqrt{4 \log(8T/\varrho)} \right)^2 \right\},$$

$$k_2 \chi^2 \geq a_2 \cdot \max \left\{ \text{rank}(X) \log^4(n), \log^{8/3}(n) g^X, \log^4(n) \log \left(\frac{4T\bar{c}_H}{\varrho} \right), 2 + \log \left(\frac{\bar{c}_{\text{coh}} T n^2}{\chi \varrho} \right) \log \left(\frac{c_{\text{coh}} T n^2}{\varrho} \right) \right\}$$

for $\chi \leq 1/4$ and constants a_0, a_2 , and recall that g^X was defined in the theorem statement and is given by $g^X := \left(\frac{\varepsilon d}{\sqrt{\log(1/\delta)}} \cdot (\lambda_{\max}^X - \lambda_{\min}^X) \right)^{2/3}$.

Now, given a vector θ , let

$$\hat{\theta}_{t+1}(\eta, \sigma; \theta) := \theta + \left(\frac{1}{k_1} (\mathbf{S}_{\text{tot},t} \bar{X}_\eta)^\top (\mathbf{S}_{\text{tot},t} \bar{X}_\eta) \right)^{-1} \left(X^\top (Y - X\theta) - \eta^2 \theta + \sigma \zeta_t \right)$$

where $\mathbf{S}_{\text{tot},t} := (\mathbf{S}_{G,t}, \xi_t) \begin{pmatrix} \mathbf{S}_{H,t} & 0_{k_2 \times d} \\ 0_{d \times n} & \mathbb{I}_d \end{pmatrix}$, $\hat{\theta}_{t+1}(\eta, \sigma, \hat{\theta}_t)$ is the update step from Line 7, and we omitted the clipping provided that the previous conditions are met. Then, using Lemma 13 and Lemma 12 the analysis presented in Lev et al. [2026, Appendix. C, Step. 3] holds and yields

$$\left\| \hat{\theta}_T(\eta, \sigma) - \theta^* \right\|_{X^\top X}^2 \leq O \left(\frac{\sigma^2 \max\{d, \log(T/\varrho)\}}{\lambda_{\min}^X + \eta^2} \right) + \left(\sqrt{2}\chi \right)^{2T} \kappa^X(\eta^2) (\lambda_{\min}^X + \eta^2) \|\theta^*\|^2 + \eta^2 \|\theta^*\|^2. \quad (25)$$

Then, substituting

$$\gamma = \gamma_1 = \Theta \left(\frac{\sqrt{k_1 T \log(1/\delta)}}{\varepsilon} \right), \quad \sigma^2 = \Theta \left(\frac{T \log(1/\delta) C^2}{\varepsilon^2} \right), \quad \hat{m}_t \leq \chi \sqrt{\lambda_{\max}^X}, \quad \tilde{\lambda}_t \leq \hat{\lambda}_t$$

$$(1 - \chi) \lambda_{\min}^X \leq \hat{\lambda}_t \leq (1 + \chi) \lambda_{\min}^X, \quad \hat{m}_t \leq \tilde{m}_t \leq \hat{m}_t + \frac{12\chi T \cdot \max\{\log(3T/\delta), \log(16T/\varrho)\}}{\varepsilon},$$

$$\hat{\lambda}_t + \eta^2 = \begin{cases} \gamma(1 + 2\tilde{m}_t) & \text{if } \eta^2 > 0 \\ \hat{\lambda}_t & \text{otherwise} \end{cases},$$

$$\lambda_{\min}^X + \eta^2 \leq \frac{\hat{\lambda}_t}{1 - \chi} + \eta^2 \leq \frac{\hat{\lambda}_t + \eta^2}{1 - \chi}, \quad \lambda_{\min}^X + \eta^2 \geq \frac{\hat{\lambda}_t}{1 + \chi} + \eta^2 \geq \frac{\hat{\lambda}_t + \eta^2}{1 + \chi}$$

in (25) and using the notation $\bar{m} := \max_{t=0,1,\dots,T-1} \tilde{m}_t$ yields

$$\left\| \hat{\theta}_T(\eta, \sigma) - \theta^* \right\|_{X^\top X}^2 \leq O \left(\frac{\sigma^2 \max\{d, \log(T/\varrho)\}}{\hat{\lambda}_t + \eta^2} \right) + \left(\sqrt{2}\chi \right)^{2T} \kappa^X(\eta^2) \left(\frac{\hat{\lambda}_t + \eta^2}{1 - \chi} \right) \|\theta^*\|^2 + \eta^2 \|\theta^*\|^2, \quad \text{for all } t = 0, 1, \dots, T-1,$$

$$\begin{aligned}
&\leq \begin{cases} O\left(\frac{\sigma^2 \max\{d, \log(T/\varrho)\}}{\gamma}\right) + (\sqrt{2}\chi)^{2T} \kappa^X(\eta^2) \frac{\gamma(1+2\bar{m})}{1-\chi} \|\theta^*\|^2 + \eta^2 \|\theta^*\|^2 & \text{if } \lambda_{\min}^X < \frac{\gamma(1+2\bar{m})}{1-\chi} \\ O\left(\frac{\sigma^2 \max\{d, \log(T/\varrho)\}}{\lambda_{\min}^X}\right) + (\sqrt{2}\chi)^{2T} \kappa^X(0) \lambda_{\min}^X \frac{1+\chi}{1-\chi} \|\theta^*\|^2 & \text{otherwise} \end{cases} \\
&\leq \begin{cases} O\left(\frac{\sqrt{T \max\{d, \log(T/\varrho), g^X\} \log(1/\delta)} (C_Y^2 + \|\theta^*\|^2) (1+2\bar{m})}{\varepsilon} \left(1 + (\sqrt{2}\chi)^{2T} \kappa^X(\eta^2) + \frac{1}{1+2\bar{m}}\right) - \lambda_{\min}^X (C_Y^2 + \|\theta^*\|^2)\right) & \text{if } \lambda_{\min}^X < \frac{\gamma(1+2\bar{m})}{1-\chi} \\ O\left(\frac{T \max\{d, \log(T/\varrho)\} \log(1/\delta) (C_Y^2 + \|\theta^*\|^2)}{\varepsilon^2 \lambda_{\min}^X} + (\sqrt{2}\chi)^{2T} \kappa^X(0) \lambda_{\min}^X \|\theta^*\|^2\right) & \text{otherwise} \end{cases} \\
&\leq \begin{cases} O\left(\frac{\sqrt{T \max\{d, \log(T/\varrho), g^X\} \log(1/\delta)} (C_Y^2 + \|\theta^*\|^2) (1+2\bar{m})}{\varepsilon} \left(1 + \frac{1}{1+2\bar{m}}\right) - \lambda_{\min}^X (C_Y^2 + \|\theta^*\|^2)\right) & \text{if } \lambda_{\min}^X < \frac{\gamma(1+2\bar{m})}{1-\chi} \\ O\left(\frac{T \max\{d, \log(T/\varrho)\} \log(1/\delta) (C_Y^2 + \|\theta^*\|^2)}{\varepsilon^2 \lambda_{\min}^X} + 8^{-T} \lambda_{\min}^X \|\theta^*\|^2\right) & \text{otherwise} \end{cases}
\end{aligned}$$

where we have used the lower bound $\hat{\lambda}_t + \eta^2 = \gamma(1 + 2\tilde{m}_t) \geq \gamma$ for the case where $\eta^2 > 0$ and the last step is for $\chi \leq \frac{1}{4}$ and $\frac{\chi \sqrt{\kappa^X(\eta^2)}}{2-\chi} < 1$. In the first case, the optimal T is constant; in the second, $T^* = O\left(\log\left(\frac{(\varepsilon \lambda_{\min}^X)^2}{\max\{d, \log(1/\varrho)\} \log(1/\delta)}\right)\right)$. Substituting this, $\bar{m} \leq O\left(\chi \left(\sqrt{\lambda_{\max}^X} + \frac{T \max\{\log(T/\delta), \log(T/\varrho)\}}{\varepsilon}\right)\right)$, and noting that the asymptotics are unchanged under $\lambda_{\min}^X < \gamma(1 + 2\bar{m})$ completes the proof. $\square$

E.2.3 Runtime Argument:

The algorithm runtime is dominated by T steps of **FastMix** (splitted between **MultipleFastMixing** and Line. 5) and T times of solving a linear regression problem of size $k_1 \times d$ (Line. 7). The complexity of the former was established in Theorem 1, and the complexity of the latter is given by $O(k_1 d^2 + d^3)$ via standard arguments. Summing these together yields

$$O\left(T \cdot (k_1 k_2 d + n d \log(n) + k_1 d^2 + k_2 d^2 + d^3)\right).$$

Substituting the previous values of k_1, k_2 and T^* yields the stated guarantee. $\square$

F Coherence for SRHT

Recall that for SRHT $\|S_{\mathbf{f}} e_i\|^2 = 1$ for all $i \in [n]$. Then, $\hat{\Delta}$ (formally defined in Line 1) is the *coherence* of $S_{\mathbf{f}}$. By Cauchy-Schwarz and the normalization of the columns of $S_{\mathbf{f}}$, it can be shown to be upper-bounded by 1. Moreover, by Welch's lower bound [Welch, 1974, Theorem 1] (see also the discussion in Haikin et al. [2021, Section 4.1]),

$$\hat{\Delta} \geq \sqrt{\frac{n-k}{k(n-1)}}.$$

In particular, when $\frac{n}{k} \gg 1$, the optimal coherence scale permitted by Welch's bound is $\Theta(k^{-1/2})$, while for $n \approx k + 1$ it might get as small as $\Theta(k^{-1})$. For the SRHT, as shown in Appendix E, w.p. at least $1 - \varrho$ there exists constants $(C, c, \bar{C})$ such that $\hat{\Delta} \leq \chi$ for every $\chi \in (0, 1]$ satisfying

$$k\chi^2 \geq C \left(2 + \log\left(\frac{cn^2}{\chi\varrho}\right) \log\left(\frac{\bar{c}n^2}{\varrho}\right)\right).$$

Hence, the resulting upper bound is of the form $\hat{\Delta} = \tilde{O}\left(\frac{\log(n/\varrho)}{\sqrt{k}}\right)$; Relative to Welch’s lower bound, the SRHT coherence bound degrades by a factor of roughly $\log(n/\varrho)$ whenever $n \gg k$.

G Algorithms

Algorithm 3 Multiple Fast Mixing

Input: Dataset $X \in \mathbb{R}^{n \times d}$, row norm bound C_X , parameters $\gamma, \tau, \omega, k, T$.

- 1: Set $\eta = 0$.
 - 2: **for** $t = 0, 1, \dots, T - 1$ **do**:
 - 3: Sample $S_{H,t} \stackrel{\text{iid}}{\sim} \text{SRHT}(k, n)$.
 - 4: Compute $\tilde{X}_t \leftarrow S_{H,t} X$.
 - 5: Compute $\hat{m}_t \leftarrow \max_{i \in [n]} \left\| \tilde{X}_t^\top S_{H,t} e_i - x_i \right\|$, $\hat{\Delta}_t \leftarrow C_X \cdot \max_{i,j: i \neq j} \left| e_i^\top S_{H,t}^\top S_{H,t} e_j \right|$, $\hat{\lambda}_t \leftarrow \lambda_{\min}^{\tilde{X}_t}$.
 - 6: Compute $\tilde{m}_t \leftarrow \max \left\{ 0, \hat{m}_t + \omega \hat{\Delta}_t (\tau - z_{1,t}) \right\}$ for $z_{1,t} \sim \text{Lap}(0, 1)$.
 - 7: Compute $\tilde{\lambda}_t \leftarrow \max \left\{ 0, \hat{\lambda}_t - \omega C_X (C_X + 2\tilde{m}_t) (\tau - z_{2,t}) \right\}$ for $z_{2,t} \sim \text{Lap}(0, 1)$.
 - 8: Set $\tilde{\eta}_t \leftarrow \sqrt{\max \left\{ 0, \gamma C_X (C_X + 2\tilde{m}_t) - \tilde{\lambda}_t \right\}}$.
 - 9: If $\tilde{\eta}_t \geq \eta$ update $\eta \leftarrow \tilde{\eta}_t$.
 - 10: **Output:** $\eta, \{\tilde{X}_t\}_{t=0}^{T-1}$.
-

H Experimental Details

The experiments were run on 12th Gen Intel(R) Core(TM) i7-1255U and implemented in Python 3.14. We used five real datasets and two synthetic datasets. The plots throughout the paper are for the normalized (divided by n) excess empirical risks, as they match the performance metric used in our accuracy theorem. In all cases, we normalized the training data so that the maximum ℓ_2 -norm of any training sample was 1 (namely, $\|x_i\|^2 \leq 1$ and $|y_i| \leq 1$ for all $i = 1, \dots, n$, so $C_X = C_Y = 1$). Moreover, to allow compatibility with the Hadamard transform, we subsample (randomly, without replacement) the datasets to a number of samples which is a power of 2. The airline dataset was subsampled to 4194304 points, to allow for reasonable solution times. For performing the Hadamard transform, we have used the optimized implementation from <https://github.com/falconn-lib/ffht>.

We report the excess empirical risk. First, we calculate the normalized mean squared error (MSE) for the train set, computed as the squared error in predicting y_i via $x_i^\top \hat{\theta}$, averaged over the train set, and we call this quantity train MSE. The excess empirical risk is defined as the difference between this quantity and the error attained by the optimal solution θ^* . The results are averaged over 30 independent trials, and we report both the empirical means and 95% confidence intervals, calculated via $\pm 1.96 \cdot \frac{\text{std}}{\sqrt{\#\text{runs}}}$. For running times, we measure wall-clock time via the function `perf_counter_ns` from Python’s `time` library, and report the average across the experiments as well as 95% confidence intervals.

Throughout the experiments, we fixed the failure probability to $\varrho = \delta/10$, so the system failure probability is only slightly larger than the additive slack in the DP definition, δ . Moreover, we set

$\delta = \frac{1}{n^2}$. For IHM, we used $k = 6 \cdot \max\{d, \log(4T/\epsilon)\}$, $T = 4$ and $C = C_Y$, following similar choices in Lev et al. [2026], where since the comparison for FastIHM is done for relatively higher range of ϵ we have increased T from 3 as was simulated in Lev et al. [2026] to $T = 4$. For FastIHM, we used $k_1 = 6 \cdot \max\{d, \log(4T/\epsilon)\}$, $T = 4$ and $C = C_Y$. For both schemes, we omitted the term $-\eta^2 \hat{\theta}_t$ from the computation of $\hat{G}_t$, as it improves performance by making the iterates converge closer to θ^* than to $\theta^*(\eta^2)$. This does not affect the privacy analysis, so we omit this term in the empirical evaluation. Whenever we used the Gaussian mechanism, we computed the noise level using the analytic, tight formula from Balle and Wang [2018].

Our synthetic datasets aim to address two edge dataset settings. The first case aims to examine the performance of our algorithm in a setting where $\lambda_{\min}^X \gg 1$, and under which, as predicted by Theorem 2, the FastIHM should perform similarly to the IHM. To that end, we generate covariates that are distributed i.i.d. and uniformly on the d -dimensional sphere. In that case, we note that

$$X^\top X = \sum_{i=1}^n x_i x_i^\top \approx \frac{n}{d} \mathbb{I}_d$$

so $\lambda_{\min}^X \approx \frac{n}{d}$, which corresponds to its maximal attainable value since for any matrix X the next bounds hold $d \cdot \lambda_{\min}^X \leq \text{Tr}(X^\top X) \leq n$. Then, we generate responses according to the linear model

$$y_i = x_i^\top \theta_0 + \sigma z_i \quad (26)$$

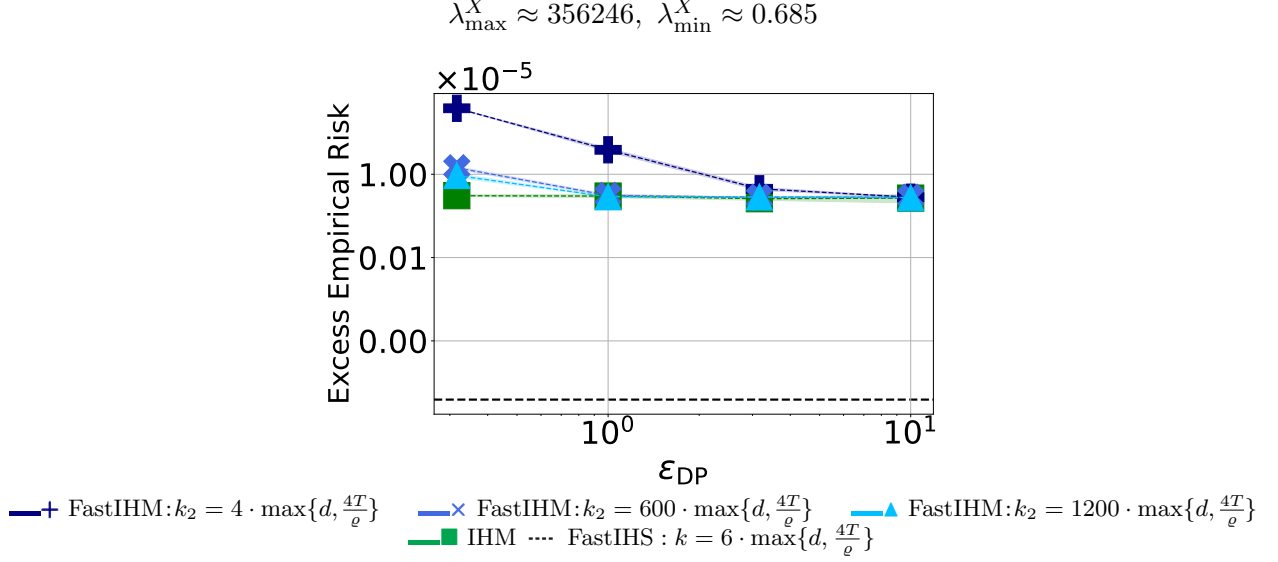
where θ_0 is a randomly generated unit vector, $z_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$ and $\sigma^2 = 0.1$. The second dataset aims to test the performance of the algorithm in a poorly conditioned setting (termed *synthetic correlated*), where usually the computational advantage of the IHS becomes prominent (see, for example, experiments done in Pilanci and Wainwright [2016], Lacotte et al. [2020]). To that end, we generate a design matrix built from vectors sampled as $x_i \stackrel{\text{iid}}{\sim} \mathcal{N}(\vec{0}_d, \Sigma)$ and where $\Sigma_{ij} = 2 \cdot (0.99)^{|i-j|}$ and responses from the same linear model (26). For both cases, we have fixed $n = 2^{19}$ and $d = 2^5$.

I Additional Experiments

We now provide additional experiments for a setting where the ratio n/d is extremely large, representing a setting where sketching is most beneficial in terms of runtime. In particular, we use the airline dataset [ope, 2020], subsampled to $n = 2^{22}$ samples, leads to $n/d = 2^{22}/4 = 2^{20}$. We sweep across values of $\frac{k_2}{\max\{d, \log(4T/\epsilon)\}} \in [4, 1200]$ and fix $T = 4, k_1 = 6 \cdot \max\{d, \log(4T/\epsilon)\}$ and $k = 6 \cdot \max\{d, \log(4T/\epsilon)\}$. As seen in Figure 4, in this regime, FastIHM allows for attaining close-to-optimal performance at a significantly lower runtime, demonstrating the most favorable regime for sketching.

J Dataset Statistics

The next table summarizes key statistics about the different datasets used for conducting the simulations in Section 4.1.



	(r_1, r_2, r_3)	Fast IHS	Fast IHM: r_1	Fast IHM: r_2	Fast IHM: r_3
Airline	(4, 600, 1200)	41.95	16.71	15.26	14.18

Figure 4: Accuracy-runtime trade-off for the Airline dataset.

Dataset	n	d	$\lambda_{\min}^X$	$\lambda_{\max}^X$	$\frac{1}{n} \ Y - X\theta^*\ ^2$	$\frac{1}{n} \ Y\ ^2$	$\frac{n}{d}$
Beijing [ruiqurm, 2018]	16384	9	$< 10^{-3}$	4435.30	0.0006	0.015	1310.7
YearMSD [Bertin-Mahieux, 2011]	131072	90	$< 10^{-3}$	464.78	0.004	0.988	1456.3
Black Friday [ope, 2019]	131072	81	$< 10^{-3}$	40022.54	0.022	0.281	1618.2
Synthetic: $\lambda_{\min}^X \approx \frac{n}{d}$	524288	32	16166.42	16637.76	0.034	0.045	16384
Synthetic: Correlated	524288	32	3.27	18723.40	0.0006	0.039	16384
Rossmann [ros, 2015]	524288	28	$< 10^{-3}$	8390.50	0.0249	0.265	18724.5
Airline [ope, 2020]	4194304	4	0.685	356246.3	0.005	0.006	1048576

Table 1: Key parameters from the datasets simulated in this work.